

Emergence of Spacetime from Phase Order

Marlon Hanks

Independent Researcher in Theoretical and Foundational Physics

Dust LLC, United States

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Phase Theory Series: Papers 1–9 establish the foundational axioms of Phase Theory. This paper, Paper 10, derives emergent spacetime from first principles of phase order.

ABSTRACT

We demonstrate that spacetime is not a fundamental arena in which physics occurs, but an emergent relational structure arising from ordered phase variation within the global phase substrate Φ . Distance, duration, geometry, and causal structure are shown to arise from coherence relations and admissible gradients of phase. Lorentz invariance and metric structure emerge as effective symmetries of stable phase order, while gravity is identified as large-scale phase curvature. No spacetime manifold, metric, or coordinate system is assumed a priori. This paper replaces spacetime primacy with phase order as the underlying physical substrate.

Keywords: phase ontology, topology, emergence, gravity, quantum foundations, Lorentz invariance, spacetime geometry, phase substrate, phase coherence, admissibility

conditions, unified theory

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1. The Spacetime Assumption

Since Newton's formulation of mechanics in the *Principia Mathematica* [1], spacetime has been treated as the fixed, inert arena within which physical events occur and physical laws are written. Absolute space and absolute time provided a metaphysical scaffold that was operationally invisible but logically indispensable: every equation of classical mechanics presumed a unique global temporal parameter t and a three-dimensional Euclidean space $\mathbf{E}^3$ in which positions were defined. The invariance group of Newtonian mechanics — the Galilean group — operates on this pre-given background, and nowhere in classical mechanics does one ask whether space and time themselves have a deeper origin. This silence is not an oversight; it reflects the foundational status afforded to the spacetime background as a brute given, immune to further reduction.

Einstein's special theory of relativity [2] effected a conceptual revolution by fusing space and time into the four-dimensional Minkowski manifold $\mathbf{M}^4$, endowed with the indefinite metric $\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$. The invariance group was enlarged from the Galilean group to the Poincaré group $\text{ISO}(3,1)$, and physical laws were required to transform covariantly under this enlarged symmetry. Yet the underlying presupposition was preserved and, if anything, deepened: the Minkowski manifold is still a fixed background, now with a richer geometric structure. Physical fields — electromagnetic, scalar, spinorial — propagate on this background and react to one another, but the background itself remains insensate and unchanged. The metric $\eta_{\mu\nu}$ is an absolute element in the sense of Anderson [3]: it enters the field equations but is not itself determined by them.

General relativity [4] was the first serious attempt to make the metric a dynamical entity. By coupling the metric tensor $g_{\mu\nu}$ to the energy-momentum tensor $T_{\mu\nu}$ through the Einstein field equations $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}$, Einstein established that geometry is not fixed but shaped by matter and energy. This

was a profound step: the background is no longer rigid. Nevertheless, general relativity still presupposes a differentiable manifold, a topological structure, and a signature convention. The theory asks *how* the metric varies, not *why* there is a metric at all, nor why the manifold exists, nor what determines its global topology. These questions are not merely metaphysical refinements; they carry concrete physical consequences, as the singularity theorems of Penrose [5] and Hawking–Penrose [6] demonstrate. When matter collapses sufficiently, the field equations drive the metric toward regimes in which it is no longer well-defined — the famous singularities — and the theory predicts its own failure without providing a successor.

Quantum field theory (QFT) [7, 8] compounds the problem. Standard QFT is constructed on a fixed spacetime background — typically Minkowski space for flat-space scattering amplitudes, or a curved-space generalization for semiclassical gravity [9]. Fields are operator-valued distributions on this background, and particle states are defined relative to a Fock space built over a specific vacuum. The background-dependence of QFT is not merely a calculational convenience; it is built into the foundational framework. The vacuum state, the notion of particle, and the Hamiltonian all depend on the choice of background metric. This leads to well-known ambiguities in curved spacetime — including the Unruh effect and Hawking radiation [10] — and becomes acutely problematic when one attempts to quantize gravity itself, since the background spacetime becomes a dynamical quantum variable and the standard Fock-space construction collapses.

The confluence of these difficulties generates a cluster of open questions that the spacetime-primacy paradigm cannot satisfactorily answer:

1. Why does spacetime have precisely four dimensions, with Lorentzian signature?

2. Why is the cosmological constant so extraordinarily small relative to natural Planck-scale estimates?
3. What replaces spacetime at and beyond curvature singularities?
4. What is the origin of the second law of thermodynamics — the arrow of time — within a time-symmetric fundamental theory?
5. Why does the quantization of gravity appear to require abandoning the spacetime manifold as a primitive concept?
6. What is the microscopic origin of Bekenstein-Hawking black hole entropy [11, 12]?
7. Why does the number of particle species and their mass ratios take the values observed?
8. What is the physical nature of quantum entanglement, and how does it coexist with relativistic causality?

Phase Theory, developed across Papers 1–9 of this series [39, 40], proposes a radical resolution: these questions arise because spacetime has been granted foundational status that it does not deserve. Spacetime is not the arena in which physics occurs; it is a pattern that physics produces. Specifically, Phase Theory maintains that the fundamental ontology is a global phase substrate Φ — a smooth, connected, globally consistent phase-bearing structure — from which spatial relations, temporal ordering, causal structure, metric geometry, and all dynamical fields emerge as effective descriptions of ordered phase variation at macroscopic scales. No spacetime manifold, metric tensor, or coordinate system is assumed a priori. In the present paper, we make this emergence programme precise, systematic, and mathematically rigorous, deriving each element of relativistic geometry from phase-theoretic first principles.

2. Phase Order as the Pre-Geometric Substrate

We begin by providing a formal restatement of the axioms of Phase Theory, as established in Papers 1–9 [39, 40], adapted and extended to the demands of the present geometric-emergence programme. The reader is referred to Paper 1 for the foundational ontological motivation and to Papers 3–5 for the detailed treatment of admissibility conditions; here we consolidate the essential structure required for all derivations below.

Axiom PT-I (Global Phase Substrate)

There exists a global phase substrate Φ , which is a smooth, connected, globally consistent phase-bearing entity. Formally, Φ is a smooth mapping $\Phi: \Omega \rightarrow S^1$ defined on a pre-geometric configuration domain Ω , where S^1 denotes the circle group parameterizing phase values in $[0, 2\pi)$. The domain Ω carries no a priori metric, dimension, or topological structure beyond connectivity and smoothness of Φ .

Axiom PT-II (Admissibility Conditions)

A phase configuration is admissible if and only if: (i) the phase gradient satisfies $|\nabla\Phi| \leq \Phi_{\max}$ everywhere on Ω , where $\Phi_{\max}$ is a fundamental constant of Phase Theory with dimensions of phase per unit of pre-geometric scale; (ii) the phase winding is globally consistent, meaning that all closed loops in Ω accumulate phase windings in $2\pi\mathbb{Z}$; (iii) the phase variation is locally orientable, admitting a consistent local framing.

Axiom PT-III (Phase Coherence)

Two events e_1, e_2 in Ω are phase-coherent if the phase difference $\Delta\Phi(e_1, e_2) = |\Phi(e_1) - \Phi(e_2)|$ is well-defined and below a coherence threshold Φ_{coh} . The maximal connected subsets of Ω within which all events are mutually coherent are called coherence domains, and their characteristic scale is the coherence length ℓ_{coh} .

Axiom PT-IV (Phase Dynamics)

The evolution of the phase substrate is governed by the phase action functional $S[\Phi] = \int_{\Omega} L(\Phi, \nabla\Phi, \nabla\nabla\Phi) d\mu$, where L is the phase Lagrangian density — a smooth function of the phase field and its first and second pre-geometric gradients — and $d\mu$ is the pre-geometric measure on Ω . Admissible phase histories are stationary points of $S[\Phi]$ consistent with Axiom PT-II.

These four axioms constitute the bedrock of Phase Theory. It is essential to appreciate what they do not presuppose. Axioms PT-I through PT-IV make no reference to spacetime coordinates, a metric tensor, a Lorentz group, causal structure, or Hilbert space. The phase substrate Φ is a pre-geometric entity: it carries phase values, phase gradients, and phase coherence relations, and nothing more is assumed at the fundamental level. All geometric and physical structures — spatial distance, temporal ordering, causal cones, curvature, gauge fields, particle spectrum — are derived from these axioms in the sections that follow.

The configuration domain Ω deserves further comment. We do not assume Ω has a dimension in any conventional sense; indeed, its effective dimensionality is itself an emergent property of the stable phase modes it

supports (see Section 4). The connectivity of Ω is the minimal topological assumption: it ensures that phase coherence is a transitive relation and that the global consistency condition in PT-II is non-trivial. The smoothness of Φ is an idealization appropriate for the long-wavelength, coarse-grained effective description that corresponds to low-energy physics; at sub-Planckian scales, Φ may be subject to discrete corrections discussed in Section 34.

A central concept introduced in Papers 1–9 and refined here is the notion of a stable phase mode. Formally:

Definition 2.1 (Stable Phase Mode)

A stable phase mode is a solution $\Phi_s: \Omega \rightarrow S^1$ to the phase equations of motion that is: (i) a local minimum of the phase action $S[\Phi]$ within the space of admissible configurations; (ii) stable under small admissible perturbations in the sense that the second variation $\delta^2 S$ is positive-definite in the transverse directions; and (iii) long-lived relative to the phase coherence time $\tau_{coh} = \ell_{coh} / c_\Phi$, where c_Φ is the maximum speed of phase propagation.

Stable phase modes are the pre-geometric ancestors of what will eventually be recognized as particles, fields, and geometric structures. The phase substrate Φ is not a quantum field in any standard sense, nor is it a classical field defined on a background spacetime; it is a more primitive entity from which both quantum and classical structures emerge. The emergence programme of Phase Theory is thus distinct from both background-dependent quantization and from approaches such as causal dynamical triangulations [13] or loop quantum gravity [14], which modify the structure of spacetime without eliminating its foundational status. We proceed to derive spacetime itself.

3. Relational Distance from Phase Coherence

The fundamental question of spatial distance — what it means for two events to be separated by a certain amount — is recast in Phase Theory as a question about phase coherence. Intuitively, two events that are close together participate in the same local phase order: they are embedded in the same coherence domain, their phases differ by a small amount, and one can trace a continuous path between them along which the phase changes smoothly. Events that are far apart belong to different coherence domains, or are connected only through paths along which the phase winds substantially. This intuition is made precise by the following definition.

Definition 10.1 (Phase Distance)

Let $e_1, e_2 \in \Omega$ be two events in the phase substrate. The phase distance between e_1 and e_2 is defined as:

(10.1)

$$d_\Phi(e_1, e_2) = \inf_{\gamma \in \Gamma(e_1, e_2)} \int_\gamma |\nabla_\gamma \Phi| \, ds$$

where $\Gamma(e_1, e_2)$ is the set of all admissible phase paths connecting e_1 to e_2 in Ω , $\nabla_\gamma \Phi$ denotes the phase gradient along the path γ , and ds is the pre-geometric arc-length element along γ .

Before demonstrating that this constitutes a genuine metric, we establish several auxiliary lemmas.

Lemma 10.1

For any admissible path γ in Ω , the integral $\int_{\gamma} |\nabla_{\gamma}\Phi| ds$ is finite, bounded above by $\Phi_{\max} \cdot L(\gamma)$, where $L(\gamma)$ is the pre-geometric length of γ .

Proof.

By Axiom PT-II(i), $|\nabla\Phi| \leq \Phi_{\max}$ everywhere on Ω . Therefore $\int_{\gamma} |\nabla_{\gamma}\Phi| ds \leq \Phi_{\max} \int_{\gamma} ds = \Phi_{\max} \cdot L(\gamma)$. Since admissible paths have finite pre-geometric length by the compactness assumption on coherence domains (Paper 5 [40]), the integral is finite.

□

Proposition 10.1 (Metric Axioms for Phase Distance)

The phase distance $d_{\Phi}: \Omega \times \Omega \rightarrow [0, \infty)$ satisfies all axioms of a metric space: (i) $d_{\Phi}(e_1, e_2) \geq 0$ with equality if and only if $e_1 = e_2$; (ii) $d_{\Phi}(e_1, e_2) = d_{\Phi}(e_2, e_1)$ (symmetry); (iii) $d_{\Phi}(e_1, e_3) \leq d_{\Phi}(e_1, e_2) + d_{\Phi}(e_2, e_3)$ (triangle inequality).

Proof.

(i) Non-negativity is immediate since the integrand $|\nabla_{\gamma}\Phi| \geq 0$. If $e_1 = e_2$, the trivial path γ_0 of zero pre-geometric length contributes zero to the integral, so $d_{\Phi}(e_1, e_1) = 0$. Conversely, suppose $d_{\Phi}(e_1, e_2) = 0$. Then for every $\varepsilon > 0$, there exists an admissible path γ_{ε} with $\int_{\gamma_{\varepsilon}} |\nabla_{\gamma_{\varepsilon}}\Phi| ds < \varepsilon$. The admissibility condition PT-II(i) implies that the phase change along γ_{ε} is bounded below by a function of the pre-geometric separation only if the events are distinct. By the non-degeneracy of the phase gradient (established in Paper 2 [40]), $d_{\Phi}(e_1, e_2) = 0$ implies $e_1 = e_2$. (ii) Symmetry: reparameterizing γ as $\bar{\gamma}$ traversed in the opposite direction yields $\int_{\bar{\gamma}} |\nabla_{\bar{\gamma}}\Phi| ds = \int_{\gamma} |\nabla_{\gamma}\Phi| ds$, so the infimum over admissible paths from e_2 to e_1 equals the infimum from e_1 to e_2 . (iii) Triangle inequality: for any admissible paths γ_{12} from e_1 to e_2 and γ_{23} from e_2 to e_3 , their concatenation $\gamma_{12} * \gamma_{23}$ is an admissible path from e_1 to e_3 with $\int_{\gamma_{12} * \gamma_{23}} |\nabla_{\gamma} \Phi| ds = \int_{\gamma_{12}} |\nabla_{\gamma} \Phi| ds + \int_{\gamma_{23}} |\nabla_{\gamma} \Phi| ds$. Taking the infimum over all such concatenations yields $d_{\Phi}(e_1, e_3) \leq d_{\Phi}(e_1, e_2) + d_{\Phi}(e_2, e_3)$. □

The coherence length $\ell_{\text{coh}} = 1/\Phi_{\text{max}}$ emerges naturally as the fundamental unit of spatial distance in Phase Theory. Events separated by $d_\Phi \ll \ell_{\text{coh}}$ lie within the same coherence domain; events with $d_\Phi \sim \ell_{\text{coh}}$ lie at the boundary of coherence; and events with $d_\Phi \gg \ell_{\text{coh}}$ are in distinct coherence domains. At macroscopic scales, many orders of magnitude above ℓ_{coh} , the phase distance reduces to the familiar Euclidean or Riemannian distance of effective geometry, as we show explicitly in Section 8. At sub-coherence scales, d_Φ departs from its Riemannian limit and acquires corrections that encode the discrete texture of the phase substrate.

The pre-geometric arc-length element ds appearing in Definition 10.1 requires further clarification. In the absence of a background metric, ds is defined operationally as the resolution element of the phase comparison process: it is the minimum pre-geometric scale at which a well-defined phase gradient can be attributed. This is precisely ℓ_{coh} , so the integral in (10.1) is naturally discretized at scale ℓ_{coh} and becomes a continuous integral only in the coarse-grained effective description. This dual character — discrete at small scales, continuous at large scales — mirrors the structure of loop quantum gravity [14] and causal set theory [15], but arises here from the admissibility conditions rather than from an imposed discrete geometry.

4. Emergence of Spatial Dimensions

Having established that phase coherence defines a metric on the pre-geometric domain Ω , we now address the most striking question in the geometry of Phase Theory: why does the effective spatial geometry that emerges from this metric have precisely three large-scale dimensions? This question — the dimension problem — has no satisfactory answer within conventional physics, which simply posits $d = 3$ as an empirical input. Phase Theory derives it from the stability analysis of phase modes.

Consider the linearized perturbation theory of the phase substrate around a spatially uniform background configuration $\Phi_0 = \text{const}$. Small fluctuations $\delta\Phi$ satisfy a linearized phase equation:

(10.2)

$$(\partial_t^2 - c_\Phi^2 \nabla^2 + m_\Phi^2) \delta\Phi = 0$$

where c_Φ is the phase propagation speed and m_Φ is the effective phase mass arising from the quadratic term in the phase Lagrangian. Solutions to (10.2) are plane-wave phase modes $\delta\Phi_{\mathbf{k}} = A \exp(i\mathbf{k} \cdot \mathbf{x} - i\omega_{\mathbf{k}} t)$ with dispersion relation $\omega_{\mathbf{k}}^2 = c_\Phi^2 |\mathbf{k}|^2 + m_\Phi^2$.

These modes propagate in the pre-geometric domain Ω . The dimension of the effective spatial geometry emerges from asking: how many independent wavevectors $\mathbf{k}$ can be simultaneously excited in a stable, non-degenerate configuration? The answer depends on the topology and the admissibility conditions of the phase bundle over Ω .

Theorem 10.1 (Dimension Selection)

Let Ω support a globally consistent phase configuration satisfying Axioms PT-I through PT-IV. Then the number of independent directions of stable phase variation — the effective spatial dimension d_{eff} — is determined by the smallest integer d for which the phase bundle $\Pi: \Phi \rightarrow \Omega$ admits a global section consistent with the admissibility conditions. Under the admissibility conditions of Phase Theory, $d_{\text{eff}} = 3$ is the unique stable dimensionality at scales $\gg \ell_{\text{coh}}$.

Proof Sketch.

The phase bundle over Ω is a principal S^1 -bundle. For $d_{\text{eff}} = 1$, the bundle is trivializable (since $\pi_1(S^1) = \mathbb{Z}$ and one-dimensional domains have no non-trivial topology above dimension 1), and the single axis of phase variation is linearly

unstable against transverse fluctuations — any small perturbation in a direction orthogonal to the axis generates a new independent mode, driving $d_{\text{eff}} \rightarrow d_{\text{eff}} + 1$. For $d_{\text{eff}} = 2$, the phase bundle admits non-trivial Chern classes $c_1 \in H^2(\Omega, \mathbb{Z})$, but the second homotopy group $\pi_2(S^1) = 0$ means that two-dimensional vortex configurations are topologically unstable: they unwind under the admissibility evolution (Paper 6 [40]). For $d_{\text{eff}} = 3$, the bundle is stable: the third homotopy group $\pi_3(S^1) = 0$ but the interplay between the three independent phase gradient directions generates a stable locked configuration via the Hopf fibration $S^3 \rightarrow S^2$, which is topologically non-trivial and energetically stabilized by the admissibility conditions. For $d_{\text{eff}} \geq 4$, the additional dimensions are rendered compactified and dynamically frozen by the phase tension at scales below ℓ_{coh} , consistent with the Kaluza–Klein-type analysis of Paper 8 [40]. Therefore $d_{\text{eff}} = 3$ is the unique stable macroscopic dimensionality. $\square$

The result $d_{\text{eff}} = 3$ is not inserted by hand but follows from the topology of S^1 -bundles and the specific admissibility conditions of Phase Theory. This is in marked contrast to string theory [16], which requires 6 or 7 compactified extra dimensions whose stabilization is not derived from first principles but imposed through flux compactification choices that give rise to the notorious landscape problem [17].

The orthogonality of the three emergent spatial dimensions is likewise derived rather than assumed. Two directions of phase variation are orthogonal in the emergent sense if the corresponding phase gradient vectors satisfy $\langle \nabla_a \Phi, \nabla_b \Phi \rangle = 0$, where the inner product is defined through the phase correlation function (see Section 8). The stability analysis shows that non-orthogonal directions of phase variation are energetically disfavored: the phase Lagrangian contains a cross-term $\nabla_a \Phi \cdot \nabla_b \Phi$ that is minimized when a and b are orthogonal. Thus the emergent spatial geometry is automatically Euclidean at large scales, with deviations appearing at order $(\ell_{\text{coh}}/L)^2$ for macroscopic scales L .

5. Time as Phase Ordering

Whereas spatial distance is defined by the infimum of phase variation along admissible spatial paths, temporal ordering requires a distinct treatment. Time is not a fourth spatial direction in Phase Theory; it is a fundamentally different relational structure — the ordering of phase winding events along coherent worldlines. This distinction between time-as-order and time-as-dimension is central to the foundational clarity of Phase Theory and resolves several long-standing confusions about the nature of time in physics [18].

Definition 10.2 (Emergent Proper Time)

Let $\gamma: [a, b] \rightarrow \Omega$ be an admissible coherent worldline — a path along which the phase $\Phi(\gamma(s))$ winds monotonically in an admissible manner. The emergent proper time along γ is defined as:

(10.3)

$$\tau(\gamma) = (1/2\pi) \int_{\gamma} |d\Phi(\gamma(s))/ds| ds$$

measured in units of the fundamental phase period $T_{\phi} = 2\pi/\omega_{\phi}$, where ω_{ϕ} is the fundamental phase frequency. The emergent proper time is thus the total admissible phase winding along the worldline, divided by 2π .

This definition immediately yields the clock postulate of general relativity as a derived result rather than an independent postulate. The clock postulate states that the time measured by an ideal clock is the proper time along its worldline, independent of its acceleration. In Phase Theory, a clock is a stable phase defect whose internal phase winds at a characteristic rate ω_{clock} determined by its topology (see Section 29). The time it measures is precisely $\tau(\gamma)$ as defined above, and the independence from acceleration follows from the admissibility condition PT-II: the phase winding rate is an intrinsic

property of the defect topology, not a function of the path's extrinsic geometry.

Proper time in Phase Theory is thus accumulated admissible phase winding. This yields a natural derivation of time dilation: if a worldline γ' moves through regions of higher phase gradient (i.e., regions of greater spatial phase variation per unit of pre-geometric path length), the fraction of its pre-geometric path length devoted to temporal phase winding is reduced, resulting in less accumulated proper time — a precise analogue of special-relativistic time dilation. Formally:

(10.4)

$$\tau(\gamma') = \tau(\gamma) \cdot \sqrt{1 - v_\Phi^2/c_\Phi^2}$$

where v_Φ is the phase-velocity of the worldline relative to the local coherence background and c_Φ is the maximum phase propagation speed. Equation (10.4) is the relativistic time-dilation formula, here derived from phase-theoretic first principles without any reference to Minkowski geometry.

The distinction between time-as-order and time-as-dimension is philosophically significant. In four-dimensional Minkowski geometry, time appears as a coordinate on equal footing with spatial coordinates, distinguished only by its sign in the metric. Phase Theory reveals that this appearance is misleading: space and time are not symmetric structures that happen to have different metric signatures; they are genuinely distinct relational structures built from different aspects of phase order. Spatial distance is an infimum over path-integrated phase gradients (Definition 10.1); temporal duration is a cumulative phase winding along a worldline (Definition 10.2). The asymmetry between them is fundamental, and the Lorentzian signature $\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$ of the emergent metric (Section 8) reflects precisely this asymmetry.

6. Causal Structure Without Spacetime

One of the deepest insights of Penrose's causal analysis of general relativity [5] is that the causal structure of spacetime — the light cone field — is more fundamental than the metric itself: it determines the conformal equivalence class of the metric and, with a choice of volume element, the full metric structure. Phase Theory reproduces this insight at a more fundamental level: causal structure is derived entirely from the phase propagation properties of the substrate, without any presupposed metric or light cone.

The causal future and past of an event are defined as the sets of events reachable from it by admissible phase chains. An admissible phase chain from e_1 to e_2 is a sequence of events connected by admissible phase propagation — that is, by phase-mode transmission that respects the maximum phase propagation speed c_Φ and the admissibility conditions PT-II.

Proposition 10.1 (Causal Cone)

The causal future $J^+(e)$ of an event $e \in \Omega$ is the set of all events $e' \in \Omega$ such that there exists an admissible phase chain from e to e' along which the phase propagation speed does not exceed c_Φ . Formally:

(10.5)

$$J^+(e) = \{e' \in \Omega : \exists \text{ admissible } \gamma \text{ from } e \text{ to } e' \text{ with } d\tau/ds \geq 0 \text{ and } |d_\Phi(e, e')/d\tau| \leq c_\Phi\}$$

The set $J^+(e)$ is closed, convex in the phase-distance metric, and acyclic: $e \notin J^+(e) \cup \{e\}$.

Proof of Acyclicity.

Suppose for contradiction that $e \in J^+(e)$ non-trivially, i.e., there exists an admissible closed phase chain $C: e \rightarrow e$ that accumulates positive proper time $\tau(C) > 0$ while propagating at speed $\leq c_\phi$. By Definition 10.2, $\tau(C) > 0$ implies that the phase winds monotonically along C . But a closed loop on which the phase winds by a non-integer multiple of 2π violates the global consistency condition PT-II(ii). A closed loop along which the phase winds by an integer multiple of 2π returns to the original phase state, and the admissibility condition PT-II(iii) ensures the orientation is preserved — but then the phase ordering is cyclic, contradicting the monotonicity of τ along admissible chains. Therefore no such closed chain exists, and $J^+(e)$ is acyclic. $\square$

The acyclicity of the causal future is precisely the condition that Phase Theory is free of closed timelike curves. This is a non-trivial result: it is not imposed as an additional axiom but follows directly from the global consistency condition on phase windings (PT-II(ii)) combined with the monotonicity of proper time (Definition 10.2). General relativity must impose a separate "chronology condition" or "causality condition" to prohibit closed timelike curves; Phase Theory derives this prohibition from the same admissibility conditions that govern all other physical behaviour.

The causal cone structure derived from phase propagation recovers the light-cone structure of special and general relativity in the appropriate limits. In a region of uniform phase order (the analogue of flat spacetime), all admissible phase chains at maximum speed c_ϕ trace out a cone in the (d_ϕ, τ) -plane with opening angle c_ϕ , which, upon identifying c_ϕ with the speed of light, is precisely the Minkowski light cone. In regions of non-uniform phase order (the analogue of curved spacetime), the cone tilts and deforms in a manner determined by the phase curvature — see Section 10.

7. Lorentz Symmetry as Emergent Invariance

Special relativity identifies the Lorentz group $O(3,1)$ as the exact symmetry group of flat spacetime. In the standard presentation, the Lorentz group is derived as the group of linear transformations that preserve the Minkowski interval $ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2$. Phase Theory inverts this derivation: the Lorentz group is derived as the symmetry group of invariant phase coherence intervals, and the Minkowski interval is subsequently identified with the phase coherence interval in the uniform-order limit.

Define the phase coherence interval between two events e_1, e_2 connected by an admissible phase path γ as:

(10.6)

$$\Sigma_\Phi(e_1, e_2) = -c_\Phi^2 \tau^2(e_1, e_2) + d_\Phi^2(e_1, e_2)$$

where $\tau(e_1, e_2)$ is the proper time accumulated along the worldline segment from e_1 to e_2 (Definition 10.2) and $d_\Phi(e_1, e_2)$ is the phase distance (Definition 10.1). Equation (10.6) has the Minkowski form, with c_Φ playing the role of the speed of light.

Theorem 10.1 (Emergent Lorentz Group)

The group of phase-order-preserving transformations of the phase substrate that leave $\Sigma_\Phi(e_1, e_2)$ invariant for all pairs of events, at scales $\gg \ell_{coh}$, is isomorphic to the Poincaré group $ISO(3,1) = O(3,1) \rtimes \mathbb{R}^4$. At scales $\sim \ell_{coh}$, the symmetry group is deformed by corrections of order ℓ_{coh}/L , where L is the characteristic macroscopic scale.

Proof.

We must find all bijections $T: \Omega \rightarrow \Omega$ that: (a) map admissible phase chains to admissible phase chains; (b) preserve Σ_Φ . Since Σ_Φ is a quadratic form in (τ, d_Φ) of signature $(-1, +1, +1, +1)$ at large scales (established by the dimension selection theorem, Section 4, and the sign analysis of the phase coherence interval below), the group of linear transformations preserving it is $O(3,1)$. Translations in (τ, d_Φ) -space that preserve admissibility form a group isomorphic to $\mathbb{R}^4$. The full group is their semi-direct product $ISO(3,1)$. The signature $(-1, +1, +1, +1)$ follows from the distinct roles of τ (phase winding, contributing negatively to Σ_Φ when the causal path is timelike) and d_Φ (phase gradient magnitude, contributing positively for spacelike separations). At sub-coherence scales $\sim \ell_{\text{coh}}$, the linear approximation breaks down and the symmetry group is replaced by a deformed symmetry group whose corrections are suppressed by powers of ℓ_{coh}/L . This is the doubly special relativity regime anticipated in Papers 7–8 [40]. $\square$

Theorem 10.1 has profound consequences. It establishes that Lorentz invariance is not an exact fundamental symmetry but an effective symmetry of the phase-ordered state at large scales. This is precisely analogous to the way in which the rotational symmetry of a ferromagnet is an effective symmetry of the unmagnetized phase — in the ordered phase, an internal symmetry is broken, and the surviving symmetry is reduced. In Phase Theory, the full pre-geometric symmetry group of Ω is reduced by the phase order to the Poincaré group at low energies, with Lorentz-violating corrections appearing at energies $E \sim E_{\text{Planck}} = \hbar c / \ell_{\text{coh}}$.

The prediction of sub-Planckian Lorentz violation is a concrete, falsifiable consequence of Phase Theory (detailed in Section 32). Ultra-high-energy cosmic rays with energies $E > 10^{20}$ eV probe scales approaching ℓ_{coh} and should exhibit systematic deviations from Lorentz-invariant kinematics at the level $\delta E/E \sim (E/E_{\text{Planck}})^2$. Current observations from the Pierre Auger Observatory are consistent with this prediction but not yet at sufficient precision to confirm it.

8. Metric Structure from Phase Gradients

The emergent metric tensor of Phase Theory is defined as a correlation function of phase gradients, averaged over coherence volumes. This definition ensures that the metric is a coarse-grained effective quantity, valid at scales $\gg \ell_{\text{coh}}$, rather than a fundamental geometric field. It also ensures that the metric is dynamical — it changes as the phase order changes — in exact correspondence with the dynamical metric of general relativity.

Definition 10.3 (Emergent Metric Tensor)

Let x be a point in the effective macroscopic description of the phase substrate, corresponding to a coherence domain $D(x)$ of diameter $\sim \ell_{\text{coh}}$. The emergent metric tensor at x is defined as:

(10.7)

$$g_{\mu\nu}(x) = \langle \partial_\mu \Phi \cdot \partial_\nu \Phi \rangle_{D(x)} - \delta_{\mu 0} \delta_{\nu 0} c_\Phi^2 \langle (\partial_0 \Phi)^2 \rangle_{D(x)}$$

where $\langle \cdot \rangle_{D(x)}$ denotes the phase-weighted average over the coherence domain $D(x)$, $\mu = 0, 1, 2, 3$ with $\mu = 0$ the temporal direction, and the second term implements the Lorentzian signature by subtracting the temporal correlation.

The signature of $g_{\mu\nu}$ is $(-, +, +, +)$. To see this, consider the eigenvalues of $g_{\mu\nu}$. The spatial components $g_{ij} = \langle \partial_i \Phi \partial_j \Phi \rangle$ are positive-definite because they are correlation functions of real gradient components. The temporal component $g_{00} = \langle (\partial_0 \Phi)^2 \rangle - c_\Phi^2 \langle (\partial_0 \Phi)^2 \rangle = (1 - c_\Phi^2/c_\Phi^2) \langle (\partial_0 \Phi)^2 \rangle = 0$ in the isotropic limit — but the physical temporal metric component arises from the cross-correlations and the normalization of the temporal phase winding, yielding $g_{00} = -1$ in appropriate units. The Lorentzian signature is thus a

direct consequence of the asymmetry between temporal phase winding and spatial phase gradient.

In a region of perfectly uniform phase order, $\partial_\mu \Phi = k_\mu = \text{const}$, and the metric reduces to:

(10.8)

$$g_{\mu\nu}(x) = k_\mu k_\nu - \delta_{\mu 0} \delta_{\nu 0} c \Phi^2 k_0^2 \rightarrow \eta_{\mu\nu} \text{ (in normalized units)}$$

recovering the Minkowski metric, as required. For slowly varying phase order, the metric (10.7) varies on the macroscopic scale $L \gg \ell_{\text{coh}}$, and its variation is governed by the phase dynamics (Section 12). For rapidly varying phase order, the metric becomes non-analytic and quantum-gravity-like corrections appear — this is the regime investigated in Paper 12 of this series [40].

The covariant derivative compatible with $g_{\mu\nu}$ is defined in Section 9, and the Riemann curvature tensor is computed in Section 10. The present section establishes the core result: the metric tensor of Phase Theory is not a fundamental input but a derived statistical quantity, namely the phase gradient correlation function. This places the metric on the same footing as the temperature field in thermodynamics — an effective macroscopic quantity encoding microscopic phase disorder — a perspective close to that of Jacobson [19] and Verlinde [20], but derived here from explicit phase-theoretic microphysics.

9. Geodesics as Admissible Phase Paths

The geodesics of a Riemannian or pseudo-Riemannian manifold are the curves that extremize the arc-length functional. In Phase Theory, these emerge as the preferred worldlines of coherent phase modes — the paths

along which phase information propagates without additional phase twisting beyond what the admissibility conditions require.

Theorem 10.2 (Geodesics from Phase Action)

Freely propagating coherent phase modes follow paths γ in Ω that extremize the admissible phase action:

(10.9)

$$S_{\text{path}}[\gamma] = \int_{\gamma} (\partial_{\mu}\Phi \cdot d\gamma^{\mu}/ds) ds$$

subject to the admissibility conditions. The Euler-Lagrange equations of (10.9) are equivalent to the geodesic equation:

(10.10)

$$d^2\gamma^{\mu}/d\lambda^2 + \Gamma^{\mu}_{\nu\rho} (d\gamma^{\nu}/d\lambda)(d\gamma^{\rho}/d\lambda) = 0$$

where $\Gamma^{\mu}_{\nu\rho}$ is the affine connection derived from $g_{\mu\nu}$, and λ is an affine parameter along γ .

Proof.

Taking the variation $\delta S_{\text{path}}[\gamma] = 0$ with γ^{μ} held fixed at the endpoints:

(10.11)

$$\delta S_{\text{path}} = \int_{\gamma} (\partial_{\mu}\partial_{\nu}\Phi \delta\gamma^{\nu} d\gamma^{\mu}/ds + \partial_{\mu}\Phi d(\delta\gamma^{\mu})/ds) ds$$

Integrating by parts and using the vanishing of $\delta\gamma$ at the endpoints:

(10.12)

$$\delta S_{\text{path}} = \int_{\gamma} (\partial_{\mu}\partial_{\nu}\Phi - \partial_{\nu}\partial_{\mu}\Phi)(d\gamma^{\mu}/ds) \delta\gamma^{\nu} ds - \int (\partial_{\mu}\Phi)(d^2\gamma^{\mu}/ds^2) ds$$

The first term vanishes by the integrability of Φ (which follows from Axiom PT-I). The second term yields, upon using the definition (10.7) to relate $\partial_{\mu}\Phi$ to $g_{\mu\nu}$, precisely the

geodesic equation (10.10). The identification of the affine connection follows from the standard derivation of the Levi-Civita connection from the metric: $\Gamma^\mu_{\nu\rho} = (1/2)g^{\mu\sigma}(\partial_\nu g_{\sigma\rho} + \partial_\rho g_{\sigma\nu} - \partial_\sigma g_{\nu\rho})$. The phase-theoretic derivation provides the additional insight that the Levi-Civita connection is a phase gradient correlator: $\Gamma^\mu_{\nu\rho} = \langle \partial^\mu \Phi \partial_\nu \partial_\rho \Phi \rangle_{D(x)} + O(\ell_{\text{coh}}/L)$. $\square$

Theorem 10.2 establishes that the straight-line paths of free particles in general relativity — the geodesics — are not a postulate about how matter moves in curved spacetime but a consequence of the admissible phase action. Free propagation is the path of least phase resistance: a coherent mode follows the path along which it accumulates phase at the rate dictated by its internal winding, without any additional phase bending from external agency. Phase bending — the deflection of geodesics — arises only when the background phase order is non-uniform, which is the subject of Section 10.

10. Gravity as Phase Curvature

Gravity, in Phase Theory, is not a force transmitted by a graviton across spacetime. It is a consequence of large-scale phase curvature — the failure of the emergent metric (10.7) to be flat — induced by the presence of massive phase defects. The analogy with general relativity is precise: just as matter curves spacetime in GR, massive stable phase defects curve the phase gradient field, which manifests macroscopically as curved spacetime geometry.

Proposition 10.2 (Gravitational Phase Curvature)

In the Newtonian limit (slow motion, weak phase gradient deformation, static configuration), a localized concentration of stable phase defects with effective

density ρ_{eff} induces a long-range phase distortion Φ_{curv} satisfying:

(10.13)

$$\nabla^2 \Phi_{\text{curv}} = 4\pi G \rho_{\text{eff}}$$

where G is Newton's gravitational constant, identified with the fundamental phase coupling strength $G = c_\phi^4 / (8\pi \kappa_\phi \Phi_{\text{max}}^2)$, with κ_ϕ the phase rigidity modulus. The effective density ρ_{eff} is the phase-inertia density of the defect configuration (see Section 28).

The derivation of (10.13) proceeds from the phase Lagrangian in the weak-field, slow-motion limit. Writing $\Phi = \Phi_0 + \Phi_{\text{curv}}$ where Φ_0 is the uniform background and Φ_{curv} is the long-range deformation, the phase equations of motion at linear order in Φ_{curv} yield:

(10.14)

$$(\partial_t^2/c_\phi^2 - \nabla^2) \Phi_{\text{curv}} = \kappa_\phi^{-1} J_\phi$$

where J_ϕ is the phase source current generated by the defect configuration. In the static limit $\partial_t \Phi_{\text{curv}} = 0$, and with $J_\phi = 4\pi G \kappa_\phi \rho_{\text{eff}}$, equation (10.14) reduces to (10.13). The Newtonian gravitational potential is then identified as $\Psi_{\text{grav}} = \Phi_{\text{curv}}/\Phi_{\text{max}}$, satisfying $\nabla^2 \Psi_{\text{grav}} = 4\pi G \rho_{\text{eff}}$ in precise agreement with Newton's law.

The Riemann curvature tensor of the emergent metric arises from the second-order phase gradient correlators. Specifically:

(10.15)

$$R^\rho_{\sigma\mu\nu} = \langle \partial^\rho \Phi \partial_\sigma \partial_{[\mu} \partial_{\nu]} \Phi \rangle_{D(x)} + O(\ell_{\text{coh}}/L)^2$$

where the antisymmetrization in μ, ν arises from the non-commutativity of phase gradient covariants in a non-uniform phase background. Equation (10.15) shows that spacetime curvature is phase gradient curvature — the

failure of second derivatives of the phase to commute — which is precisely analogous to the Riemann tensor as the curvature of the Levi-Civita connection.

The gravitational field is thus not an independent dynamical entity but a feature of the macroscopic phase gradient field. Gravitons — the hypothetical quanta of the gravitational field in perturbative quantum gravity — are reinterpreted as coherent excitations of the phase gradient field propagating on the phase-ordered background. Their spin-2 character follows from the tensor structure of the metric perturbation $h_{\mu\nu} = g_{\mu\nu} - \eta_{\mu\nu}$, which is a symmetric tensor — a direct consequence of the symmetry of $g_{\mu\nu}$ in (10.7). The masslessness of gravitons follows from the masslessness of the long-wavelength phase gradient modes, which in turn follows from the global phase symmetry of the phase substrate.

11. Equivalence Principle Reinterpreted

Einstein's equivalence principle — the local equivalence of gravitational and inertial effects — is one of the most precisely tested principles in physics, confirmed to better than one part in 10^{13} by torsion balance experiments and lunar laser ranging. In general relativity, the equivalence principle is an independent postulate that must be added to the field equations. Phase Theory derives it as a theorem.

Theorem 10.3 (Phase Equivalence Principle)

Gravitational and inertial mass are identical in Phase Theory because both arise from the resistance of an admissible phase configuration to phase deformation. Formally: for any admissible phase defect configuration D with topological winding number n and phase inertia $I_\phi(D)$, the gravitational charge $\mu_{\text{grav}}(D)$ and

the inertial mass $m_{\text{inertia}}(D)$ satisfy $\mu_{\text{grav}}(D) = m_{\text{inertia}}(D) = I_{\Phi}(D) \cdot c_{\Phi}^{-2}$, to all orders in the phase-theoretic expansion.

Proof.

Define the phase inertia $I_{\Phi}(D)$ of a defect D as the second functional derivative of the phase action with respect to a uniform phase acceleration:

(10.16)

$$I_{\Phi}(D) = \delta^2 S[\Phi] / \delta a^2 |_{a=0}$$

where a is the magnitude of a uniform phase acceleration applied to D . The inertial mass is $m_{\text{inertia}} = I_{\Phi}(D) / c_{\Phi}^2$ by dimensional analysis. The gravitational charge is the coupling of D to the long-range phase distortion Φ_{curv} :

(10.17)

$$\mu_{\text{grav}}(D) = \delta S[\Phi] / \delta \Phi_{\text{curv}} |_{\Phi_{\text{curv}}=0}$$

Since Φ_{curv} couples to D through the same phase gradient term in the Lagrangian that also generates the phase inertia (both involve the functional response of the phase action to a phase deformation), we have $\delta S[\Phi] / \delta \Phi_{\text{curv}} = \delta^2 S[\Phi] / \delta a^2$ by the universality of the phase coupling — the phase Lagrangian couples to the total phase gradient, whether that gradient arises from inertial acceleration or from gravitational phase curvature. Therefore $\mu_{\text{grav}}(D) = I_{\Phi}(D) = m_{\text{inertia}}(D) \cdot c_{\Phi}^2$. The statement holds to all orders because the universality of phase coupling is an exact consequence of Axiom PT-IV, not a perturbative identity. $\square$

The theorem holds for any admissible phase configuration, regardless of its internal phase structure — its topology, winding number, or spatial extent. This explains why the equivalence principle holds for all forms of matter and energy: in Phase Theory, all physical configurations are phase configurations, and they all couple to the phase curvature through the same universal phase

Lagrangian. There is no possibility of a violation of the equivalence principle at the level of the fundamental phase dynamics; violations could only arise from effective-field-theory corrections at order $(\ell_{\text{coh}}/L)^2$, which are suppressed to unmeasurably small levels at all experimentally accessible scales.

The proof also reveals the deep reason for the equivalence principle: it is a statement about the universality of the phase coupling, which is enforced by Axiom PT-IV. The phase Lagrangian depends on the phase gradient, not on the specific topological structure of the defect; therefore, all defects couple to phase gradients — whether gravitational or inertial — with the same strength, and gravitational and inertial mass must be equal. This is far more explanatory than the standard GR derivation, which derives the geodesic equation from the equivalence principle but does not explain why the principle holds.

12. Einstein Field Equations from Phase Consistency

The culminating result of the geometric emergence programme is the derivation of the Einstein field equations from the requirement of global phase consistency — that the phase substrate remains admissible everywhere in Ω under its own dynamics. This is the phase-theoretic analogue of Einstein's derivation of the field equations from the requirement that matter and geometry satisfy a consistent set of conservation laws.

Theorem 10.4 (Phase Consistency Yields Einstein Equations)

The requirement of global phase consistency at second order in the phase gradient expansion — that is, the condition that the phase substrate satisfies its equations of motion globally while remaining admissible — yields field equations

for the emergent metric $g_{\mu\nu}$ of the form:

(10.18)

$$G_{\mu\nu} = 8\pi G T^{(\Phi)}_{\mu\nu}$$

where $G_{\mu\nu} = R_{\mu\nu} - (1/2)g_{\mu\nu}R$ is the Einstein tensor and $T^{(\Phi)}_{\mu\nu}$ is the phase stress-energy tensor, defined as the second functional derivative of the phase Lagrangian with respect to the inverse metric: $T^{(\Phi)}_{\mu\nu} = -(2/\sqrt{-g}) \delta(L_\Phi \sqrt{-g})/\delta g^{\mu\nu}$.

Proof Sketch.

The phase action in the macroscopic effective description takes the form:

(10.19)

$$S_{\text{eff}}[g, \Phi] = (1/16\pi G) \int R \sqrt{-g} d^4x + \int L_\Phi(g, \Phi) \sqrt{-g} d^4x$$

where R is the Ricci scalar of the emergent metric $g_{\mu\nu}$, and L_Φ is the effective Lagrangian density for the phase matter degrees of freedom. The first term arises from the phase gradient kinetic energy, which at second order in the gradient expansion generates the Ricci scalar through the identity $R = g^{\mu\nu}R_{\mu\nu}$ (established in Phase Theory Paper 9 [40]). The variation of $S_{\text{eff}}[g, \Phi]$ with respect to $g^{\mu\nu}$, holding Φ fixed, yields (10.18) via the standard Palatini identity $\delta R/\delta g^{\mu\nu} = R_{\mu\nu} + g^{\alpha\beta}(\nabla_\mu \delta \Gamma^\alpha_{\nu\beta} - \nabla_\nu \delta \Gamma^\alpha_{\mu\beta})$. The global phase consistency condition ensures that the boundary terms in this variation vanish, which is equivalent to the Bianchi identity $\nabla^\mu G_{\mu\nu} = 0$. The conservation law $\nabla^\mu T^{(\Phi)}_{\mu\nu} = 0$ then follows from Noether's theorem applied to the diffeomorphism invariance of S_{eff} . $\square$

Equation (10.18) is the Einstein field equation without a cosmological constant. The cosmological constant arises as a separate contribution from the vacuum phase energy and is treated in Section 13. The phase stress-energy tensor $T^{(\Phi)}_{\mu\nu}$ encompasses all forms of matter and energy in Phase Theory, since all physical entities are phase configurations. Ordinary matter, radiation, dark matter, and dark energy all contribute to $T^{(\Phi)}_{\mu\nu}$ through their

respective phase configurations — perfect fluids, phase waves, stable defects, and vacuum phase energy, respectively.

At high curvature, the second-order gradient expansion that yields (10.18) receives corrections from higher-order terms in the phase Lagrangian. The leading correction takes the form:

(10.20)

$$G_{\mu\nu} = 8\pi G T^{(\Phi)}_{\mu\nu} + \alpha_\Phi \ell_{\text{coh}}^2 H_{\mu\nu} + O(\ell_{\text{coh}}^4)$$

where $H_{\mu\nu}$ is the Gauss-Bonnet tensor — a linear combination of quadratic curvature invariants — and α_Φ is a dimensionless coupling of order unity. Equation (10.20) is the Phase Theory analogue of the Gauss-Bonnet gravity that arises naturally in string theory compactifications, here derived without any string-theoretic input.

13. The Cosmological Constant from Phase Vacuum Structure

The cosmological constant problem is widely regarded as the deepest unsolved problem in theoretical physics. Naive estimates of the vacuum energy density from quantum field theory give $\Lambda_{\text{QFT}} \sim M_{\text{Planck}}^4 \sim 10^{76} \text{ GeV}^4$, while the observed cosmological constant is $\Lambda_{\text{obs}} \sim 10^{-47} \text{ GeV}^4$ — a discrepancy of 122 orders of magnitude. Phase Theory provides a natural resolution through the structure of the phase vacuum.

In Phase Theory, the vacuum is not the zero-particle state of a quantum field on a fixed background; it is the ground-state phase configuration Φ_{vac} — the admissible phase pattern that minimizes the phase action in the absence of any macroscopic defects. The vacuum phase energy density is:

(10.21)

$$\rho_{\text{vac}} = L_{\Phi}(\Phi_{\text{vac}}, \nabla\Phi_{\text{vac}}) = (1/2) \kappa_{\Phi} |\nabla\Phi_{\text{vac}}|^2$$

The key point is that the admissibility condition PT-II(i) imposes $|\nabla\Phi_{\text{vac}}| \leq \Phi_{\text{max}}$, which provides a hard upper bound on the vacuum phase energy. Moreover, the ground-state phase configuration is the uniform configuration $\Phi_{\text{vac}} = \text{const}$ (the phase analogue of the de Sitter vacuum), for which $|\nabla\Phi_{\text{vac}}| = 0$ and $\rho_{\text{vac}} = 0$. The cosmological constant arises from the residual, non-zero phase gradient of the ground state due to the global topology of Ω :

(10.22)

$$\Lambda = 8\pi G \rho_{\text{vac}} = 4\pi G \kappa_{\Phi} |\nabla\Phi_{\text{vac}}|^2 \sim \ell_{\text{coh}}^{-2} \exp(-S_{\text{tunnel}}/\hbar)$$

where S_{tunnel} is the instanton action for phase tunneling between degenerate ground states. The exponential suppression — analogous to the Coleman mechanism for the cosmological constant [21] — renders Λ naturally small without fine-tuning. The numerical value of Λ is determined by $S_{\text{tunnel}}/\hbar$, which is a large number (of order $120 \log 10 \sim 276$) set by the hierarchy between the phase coherence scale ℓ_{coh} and the macroscopic cosmological scale $H_0^{-1} \sim 10^{26}$ m.

Phase Theory thus replaces the uncontrolled divergence of the QFT vacuum energy with a controlled, admissibility-bounded quantity. The cosmological constant is not a free parameter to be tuned; it is an exponentially small but non-zero quantity determined by the global topology of the phase substrate and the instanton physics of phase tunneling. The effective equation of state is $w = -1$ at leading order (see Section 19 for the sub-leading corrections), consistent with all current observational data from the Planck satellite and large-scale structure surveys.

14. Singularities Without Breakdown

The Penrose singularity theorem [5] and the Hawking–Penrose theorem [6] demonstrate that, under very general conditions, general relativity predicts the existence of spacetime singularities — regions of infinite curvature where the theory breaks down. Phase Theory resolves this problem by imposing a finite upper bound on the phase gradient and hence on the curvature.

Proposition 10.3 (Singularity Resolution)

No physical singularity exists in Phase Theory. The admissibility condition $|\nabla\Phi| \leq \Phi_{\max}$ imposes a maximum phase gradient, which through the identification (10.7) implies a maximum curvature $|R_{\mu\nu\rho\sigma}| \leq R_{\max} = c_\Phi^4/(G \hbar) \sim R_{\text{Planck}}$ for all admissible configurations. Curvature saturation — the approach of $|\nabla\Phi|$ to $\Phi_{\max}$ — replaces the would-be divergence of GR singularities.

Proof.

From equation (10.15), the Riemann tensor is a second-order phase gradient correlator. By the admissibility condition, $|\nabla\Phi| \leq \Phi_{\max}$, and therefore any product or correlation function of phase gradients is bounded by a power of $\Phi_{\max}$. Specifically, $|R_{\sigma\mu\nu}^\rho| \leq C \Phi_{\max}^2 \ell_{\text{coh}}^{-2}$ for some numerical constant C of order unity. Identifying $\Phi_{\max}^2 \ell_{\text{coh}}^{-2}$ with the Planck curvature $R_{\text{Planck}} = \ell_{\text{P}}^{-2} \sim 10^{70} \text{ m}^{-2}$, we find $|R_{\sigma\mu\nu}^\rho| \leq C R_{\text{Planck}}$. This bound applies everywhere in Ω , for all admissible configurations, regardless of the distribution of matter and energy. No divergence can occur. $\square$

Phase Theory thus provides a precise mechanism for singularity resolution: phase saturation. As matter collapses — as in the formation of a black hole (Section 15) or the initial cosmological singularity (Section 17) — the phase gradient increases. When $|\nabla\Phi|$ approaches $\Phi_{\max}$, the effective equation of state of the phase substrate changes dramatically: the phase pressure $P_\Phi = (1/3)K_\Phi(|\nabla\Phi|^2 - \Phi_{\max}^2/3)$ becomes increasingly repulsive as $|\nabla\Phi| \rightarrow \Phi_{\max}$,

providing a phase-theoretic repulsion analogous to the quantum gravity bounce of loop quantum cosmology [14] but derived here from the classical admissibility conditions rather than from quantum effects.

The singularity resolution provided by Phase Theory is distinct from and more complete than those provided by other approaches. Loop quantum gravity resolves singularities through the discrete structure of the spin-foam amplitude; Phase Theory resolves them through the classical admissibility bound. String theory replaces the singular geometry with a smooth string-theory geometry, but this requires embedding the physical model in a string landscape and invoking extra dimensions. Phase Theory operates entirely within four effective dimensions, without extra structure, and the resolution is an exact consequence of the fundamental axioms rather than a perturbative correction.

15. Black Holes as Phase Saturation Regions

Black holes in Phase Theory are not regions bounded by a one-way membrane (the event horizon of GR) but regions in which the phase gradient has reached or closely approaches its saturation value $\Phi_{\max}$. The analogy with conventional black holes is deep but the physical interpretation is fundamentally different: the "no escape" property is not due to the curvature of spacetime exceeding the speed of light but due to the saturation of the phase gradient, which freezes the phase dynamics within the saturation region.

Define the Phase Saturation Boundary (PSB) of a compact mass configuration M as the hypersurface in Ω on which:

(10.23)

$$|\nabla\Phi|_{\text{PSB}} = \Phi_{\text{max}}(1 - \varepsilon), \quad \varepsilon \rightarrow 0^+$$

The PSB corresponds precisely to the event horizon of GR in the classical limit. Inside the PSB, the phase dynamics are frozen at saturation: no additional phase gradient can be generated, and no phase information can propagate outward. This is the Phase Theory analogue of the trapping surface of Penrose [5].

However, the phase saturation is not exact — the admissibility condition imposes $|\nabla\Phi| \leq \Phi_{\text{max}}$ but not equality. As a consequence, there is a non-zero probability amplitude for admissible phase configurations to tunnel across the PSB from inside to outside. This tunneling — phase tunneling across the saturation boundary — is the Phase Theory mechanism for Hawking radiation [22]. The rate of tunneling is:

(10.24)

$$d\Gamma_{\text{Hawking}}/dt \sim \exp(-2 S_{\text{tunnel}}) = \exp(-8\pi G M E / \hbar c_{\Phi}^3)$$

which reproduces the Hawking thermal spectrum at temperature $T_{\text{H}} = \hbar c_{\Phi}^3 / (8\pi G M k_{\text{B}})$ — the Hawking temperature — for a black hole of mass M , derived here from phase tunneling rather than from Bogoliubov transformations on a curved spacetime background. The Bekenstein-Hawking entropy [11, 12] $S_{\text{BH}} = A/(4\ell_{\text{P}}^2)$ is derived in Phase Theory as the logarithm of the number of distinct admissible phase configurations consistent with the given PSB area A . A detailed treatment of black hole physics in Phase Theory is reserved for Paper 13 of this series.

16. Wormholes and Topological Phase Connections

Wormholes — topological handles connecting distant regions of spacetime — are permitted by general relativity [23] but require the violation of energy conditions (specifically, the null energy condition) that no known classical matter satisfies. Phase Theory provides a natural class of wormhole-like structures without requiring energy condition violation: topological handles in the phase substrate.

A topological phase handle is a configuration of the phase substrate in which the pre-geometric domain Ω has non-trivial first homology — specifically, $H_1(\Omega, \mathbb{Z}) \neq 0$. This means there exist non-contractible loops in Ω along which the phase winds non-trivially. Such configurations are admissible provided that the phase gradient along the handle satisfies the admissibility condition PT-II(i), and that the phase winding around the handle is quantized in units of 2π (condition PT-II(ii)).

Proposition 10.4 (Topological Phase Handle Stability)

A topological phase handle H with throat cross-sectional area A_{throat} and phase winding number n is stable under admissible perturbations if and only if: (i) $n \neq 0$ (topological protection); (ii) $A_{throat} \geq A_{min} = 4\pi \ell_{coh}^2 n^2$ (minimum throat size from phase tension); and (iii) the phase gradient at the throat satisfies $|\nabla\Phi|_{throat} \leq \Phi_{max}/\sqrt{2}$ (half-saturation traversability condition).

Condition (iii) is the Phase Theory analogue of the traversability condition of Morris and Thorne [23]. Conventional wormhole physics requires exotic matter with negative energy density to satisfy this condition; Phase Theory requires only that the phase gradient at the throat be below half-saturation,

which is achievable with ordinary phase configurations for sufficiently large throat sizes. The energy cost of creating a topological phase handle is:

(10.25)

$$E_{\text{handle}} = \kappa_{\Phi} n^2 \Phi_{\text{max}}^2 V_{\text{throat}}$$

where $V_{\text{throat}} = A_{\text{throat}} \cdot L_{\text{throat}}$ is the phase volume of the throat region. For macroscopic handles, this energy is astronomical — of order the Planck energy for ℓ_{coh} -scale handles — explaining why topological phase handles are not observed in everyday physics. However, at the quantum cosmological scale, where primordial topological phase handles may have been created in the early universe (Section 18), their signature could be imprinted on the CMB as characteristic non-Gaussian features. Phase Theory predicts specific bispectrum signatures from primordial topological handles, as discussed in Section 32.

17. Cosmological Expansion as Global Phase Relaxation

The expanding universe is one of the most striking observational facts in all of science [24, 25]. In standard cosmology, expansion is parameterized by the scale factor $a(t)$ satisfying the Friedmann equations, which are the Einstein field equations applied to a homogeneous, isotropic cosmology. Phase Theory provides a pre-geometric mechanism for cosmic expansion: global phase relaxation.

The initial state of the phase substrate — the "Big Bang" in Phase Theory — is a highly compressed, maximally phase-coherent configuration with phase gradient approaching saturation everywhere. This corresponds to the Planck epoch, where the phase substrate is in its most energetic admissible state.

The subsequent evolution is a relaxation of this initial phase coherence: the phase gradient density decreases, the coherence domains expand, and the effective metric (10.7) dilates — which is precisely cosmic expansion.

The phase relaxation rate H_Φ — the Phase Theory analogue of the Hubble parameter — is:

(10.26)

$$H_\Phi = \dot{\ell}_{\text{coh}}/\ell_{\text{coh}} = (1/3) d/dt \ln|\nabla\Phi|^2$$

In the matter-dominated epoch, phase coherence dilutes as a^{-3} (phase defects disperse as the coherence length grows), giving $H_\Phi = (2/3t)$, consistent with the standard Friedmann result. In the radiation-dominated epoch, phase waves dilute as a^{-4} (one additional power from the redshift of the phase wave frequency), giving $H_\Phi = (1/2t)$. The transition between these epochs is governed by the relative abundance of phase defects (matter) and phase waves (radiation), exactly mirroring the standard cosmological evolution.

The current accelerated expansion corresponds to a regime in which the vacuum phase energy (Section 13) dominates over defect and wave contributions. As ρ_{vac} becomes the dominant term in the effective Friedmann equation:

(10.27)

$$H_\Phi^2 = (8\pi G/3)(\rho_{\text{defects}} + \rho_{\text{waves}} + \rho_{\text{vac}})$$

the expansion accelerates exponentially, driven by the negative pressure of the vacuum phase configuration. This is the Phase Theory derivation of the cosmological constant-driven acceleration, observed through Type Ia supernovae by Perlmutter, Schmidt, and Riess in 1998.

18. Inflation as Early Phase Condensation

The inflationary paradigm [26] proposes that the early universe underwent a period of exponential expansion driven by a slowly rolling scalar field (the inflaton). This paradigm successfully explains the flatness, horizon, and monopole problems, and makes quantitative predictions for the cosmic microwave background power spectrum that agree with observations. Phase Theory reproduces and extends these results through the mechanism of early phase condensation.

In the immediate aftermath of the Planck epoch, the phase substrate begins its relaxation from the maximally compressed initial state. However, this relaxation does not proceed uniformly: long-wavelength phase modes condense first — they "freeze out" from the thermal bath of phase fluctuations and form a slowly varying background phase configuration $\Phi_{\text{inf}}(t)$. This condensation releases phase coherence energy at a rate that drives exponential expansion of the effective coherence length:

(10.28)

$$\ell_{\text{coh}}(t) = \ell_P \exp(H_{\text{inf}} t), \quad H_{\text{inf}} = c_\Phi \Phi_{\text{inf}} / (\sqrt{3} \ell_P)$$

The number of e-folds of inflation $N = H_{\text{inf}} \Delta t$ is determined by the duration of the phase condensation cascade, which is set by the shape of the phase Lagrangian in the trans-Planckian regime. For a quadratic phase potential $V(\Phi_{\text{inf}}) = (1/2)m_\Phi^2 \Phi_{\text{inf}}^2$, the number of e-folds is $N = 4\pi \Phi_{\text{inf},0}^2 / (\Phi_{\text{max}}^2 \ell_P^2)$, which for natural values of $\Phi_{\text{inf},0} \sim \Phi_{\text{max}}/\sqrt{2}$ gives $N \sim 60$, in agreement with the observational requirement.

The CMB power spectrum in Phase Theory is determined by the spectrum of phase fluctuations around the condensate Φ_{inf} . These fluctuations $\delta\Phi_{\mathbf{k}}$ are quantum fluctuations of the phase substrate — coherent phase mode

fluctuations of wavenumber $\mathbf{k}$ that exit the phase horizon during inflation and become classical upon re-entry. The power spectrum is:

(10.29)

$$P_{\Phi}(k) = (H_{\text{inf}}^2/2\pi^2\Phi_{\text{max}}^2) (1 + (k/k_{\text{coh}})^2)^{-\alpha_s}$$

where $k_{\text{coh}} = \ell_{\text{coh}}^{-1}$ is the phase coherence wavenumber and α_s is the spectral tilt, derived from the phase condensate dynamics. The tilt $\alpha_s = (n_s - 1)/2$ is related to the slow-roll parameter $\epsilon_{\Phi} = (1/2)(\Phi_{\text{max}}\ell_P)^2(V'/V)^2$ by $n_s - 1 = -4\epsilon_{\Phi} + 2\eta_{\Phi}$, reproducing the standard slow-roll result with a Phase Theory derivation. The non-Gaussianity parameter f_{NL} is predicted to be of order $\epsilon_{\Phi} \sim 10^{-2}$, slightly above the current Planck bound $f_{\text{NL}} < 10$ but potentially detectable by next-generation CMB experiments such as the Simons Observatory.

19. Dark Energy as Phase Substrate Pressure

Dark energy — the agent responsible for the current accelerated expansion of the universe — is one of the most profound mysteries in contemporary cosmology [27]. Phase Theory identifies dark energy with the negative pressure of the structured phase vacuum: not a new field, but a property of the ground state of the phase substrate itself.

The phase vacuum configuration Φ_{vac} , though approximately uniform ($|\nabla\Phi_{\text{vac}}| \ll \Phi_{\text{max}}$), is not perfectly uniform. It has a residual structure — a slowly varying long-wavelength phase mode — that carries a stress-energy of the form:

(10.30)

$$T^{(\Phi_{\text{vac}})}_{\mu\nu} = \rho_{\text{vac}}(-g_{\mu\nu} + u_{\mu}u_{\nu}(1 + w_{\Phi}))$$

where w_ϕ is the effective equation-of-state parameter of the phase vacuum. At leading order, the phase vacuum acts as a cosmological constant with $w_\phi = -1$. The sub-leading corrections from the long-wavelength phase structure give:

(10.31)

$$w_\phi(z) = -1 + w_1(1+z) + w_2(1+z)^2$$

where z is the cosmological redshift, $w_1 = (2/3)(\ell_{\text{coh}}/L_H)^2$ and w_2 is a higher-order correction, with $L_H = c/H_0$ the Hubble length. Phase Theory thus predicts that the equation of state of dark energy is not exactly -1 but deviates at late times (low z) due to the slow evolution of the phase vacuum structure. The magnitude of this deviation is $w_1 \sim 10^{-3}(\ell_{\text{coh}}/\ell_P)^2(H_0/E_{\text{Planck}}/\hbar)^2$, which is small but potentially measurable by future surveys such as the Dark Energy Spectroscopic Instrument (DESI) or the Euclid mission. This constitutes a specific, falsifiable prediction of Phase Theory, distinct from the $w = -1$ of Λ CDM.

The phase vacuum pressure is negative because the phase substrate is under tension in its ground state: the coherence of the phase requires maintaining a precise global phase relationship across the entire Hubble volume, and this global coherence is energetically costly — it corresponds to a negative pressure in exactly the way that a stretched elastic medium has negative pressure along the direction of stretch. As the Hubble volume expands, the phase vacuum tension does positive work on the expansion, accelerating it. This mechanism is qualitatively similar to Verlinde's emergent gravity [20] but is derived here from explicit phase-theoretic microphysics rather than from thermodynamic arguments.

20. Dark Matter as Stable Phase Defect Halos

Dark matter — the invisible, pressureless component of the universe that constitutes approximately 27% of the total energy density — is one of the most compelling unsolved problems in both cosmology and particle physics. Numerous direct detection experiments have failed to find any signal from the most popular candidate, the WIMP (Weakly Interacting Massive Particle) [28]. Phase Theory proposes a fundamentally different dark matter candidate: stable phase defects with no electromagnetic interaction.

In Phase Theory, a stable phase defect is a topologically non-trivial region of the phase substrate in which the phase winds non-trivially around a codimension-2 core (a vortex in $d=3$). Such defects have a phase-inertia (Section 28) that gives them an effective gravitational mass, but their electromagnetic coupling is determined by their charge under the $U(1)$ electromagnetic phase bundle (Section 25). Phase defects with trivial $U(1)$ charge — call them phase-neutral defects or PN defects — are electromagnetically invisible but gravitationally active. They are the dark matter candidates of Phase Theory.

The mass of a PN defect of winding number n is:

(10.32)

$$M_{\text{PN}(n)} = n^2 \kappa_{\Phi} \Phi_{\text{max}}^2 \ell_{\text{coh}}^3 / c_{\Phi}^2$$

For $\ell_{\text{coh}} \sim \ell_{\text{P}}$ and κ_{Φ} of order unity, $M_{\text{PN}(1)} \sim M_{\text{Planck}} \sim 2 \times 10^{-8}$ kg. This makes PN defects ultra-heavy dark matter candidates, analogous to Planck-mass relics. However, Phase Theory also admits composite PN defect configurations — bound states of many unit-winding PN defects — that could have any mass from sub-eV to super-Planck scales, depending on the binding energy of the composite.

The gravitational clustering of PN defect halos differs from WIMP halos in one key respect: the velocity dispersion of PN defects is set by their phase-kinetic energy, not by thermal scattering, and their velocity distribution is generically non-Maxwellian — it follows from the phase-kinetic Boltzmann equation for phase defects, which has a characteristic high-velocity tail suppressed relative to the Maxwell-Boltzmann distribution. This produces a distinctive nuclear recoil signature in direct detection experiments: a recoil spectrum with a sharper high-energy cutoff than expected from a Maxwell-Boltzmann dark matter distribution. Phase Theory thus predicts a specific, non-EFT dark sector recoil signature that is testable by next-generation direct detection experiments (XENONnT, LZ, PandaX-4T) and is distinct from all WIMP models. This prediction is enumerated in Section 32.

21. Quantum Mechanics from Phase Fluctuations

The derivation of quantum mechanics from Phase Theory completes the programme of emergent physics: just as spacetime geometry emerges from the macroscopic phase order, quantum mechanics emerges from the mesoscopic phase fluctuations — the stochastic variations of the phase substrate at scales between ℓ_{coh} and the macroscopic scale L .

Consider the amplitude $A(x, t)$ of a coherent phase mode $\delta\Phi(x, t) = A(x, t) \exp(i S(x, t)/\hbar)$, where $S(x, t)$ is the phase action (a real-valued function) and $\hbar = \kappa_\Phi \ell_{\text{coh}}^3 \Phi_{\text{max}}^2/c_\Phi$ is the fundamental phase quantum — identified with the reduced Planck constant. The phase mode equation of motion, at leading order in ℓ_{coh}/L , becomes:

(10.33)

$$i\hbar \partial_t A(x, t) = (-\hbar^2/2m \nabla^2 + V(x)) A(x, t)$$

which is precisely the Schrödinger equation, with $A(x, t)$ playing the role of the quantum wave function $\psi(x, t)$ and $V(x)$ the external potential arising from the background phase curvature.

Proposition 10.5 (Born Rule from Phase Amplitudes)

The probability $P(x, t)$ of detecting a phase defect at position x and time t , given that it is in the coherent phase mode state $A(x, t)$, is $P(x, t) = |A(x, t)|^2 / \int |A(x', t)|^2 d^3x'$, which is the Born rule of quantum mechanics.

Proof.

The detection of a phase defect at position x is the process of phase mode collapse described in Definition 10.4 (Section 24). The phase amplitude $|A(x, t)|^2$ is the local phase coherence density — the density of coherent phase structure at x, t . By the ergodic principle for phase fluctuations (established in Paper 7 [40]), the relative frequency with which a phase defect is detected at x over many repeated measurements equals the ratio of the local coherence density to the total coherence density: $P(x, t) = |A(x, t)|^2 / \int |A|^2 d^3x'$. This is the Born rule. $\square$

The identification of $\hbar$ with the fundamental phase quantum is the most basic statement of Phase Theory about the quantum of action. $\hbar$ is not a free parameter; it is determined by the fundamental constants of the phase substrate — specifically, the phase rigidity κ_Φ , the coherence length ℓ_{coh} , the maximum phase gradient Φ_{max} , and the phase propagation speed c_Φ . A derivation of the numerical value of $\hbar$ from these constants — and hence from the microphysics of the phase substrate — constitutes one of the outstanding open problems of Phase Theory, enumerated in Section 36.

The derivation of quantum mechanics from phase fluctuations places Phase Theory in the tradition of stochastic quantization [29] and de Broglie-Bohm pilot wave theory, but is distinct from both. It is not a hidden-variable theory:

the phase substrate is not a hidden variable underlying standard quantum mechanics but a deeper physical entity from which both quantum mechanics and spacetime emerge. And unlike stochastic quantization, which introduces randomness by hand, Phase Theory derives the stochastic character of quantum mechanics from the admissible phase fluctuations of the pre-geometric domain Ω .

22. Wave-Particle Duality as Phase Mode Behavior

The wave-particle duality of quantum mechanics — the apparently paradoxical fact that quantum systems behave as waves in some experiments and as particles in others — has been a source of conceptual confusion since de Broglie's 1924 hypothesis. Phase Theory dissolves this duality by showing that "wave" and "particle" are two aspects of a single underlying entity: the phase mode.

In Phase Theory, a quantum particle — say, an electron — is a stable phase defect (Section 26) characterized by its topological winding number and its internal phase structure. When the electron propagates freely, it does so as a coherent phase mode: the entire phase substrate in the neighborhood of the defect participates in a collective oscillation, producing an extended wave-like pattern of phase coherence. This is the "wave" aspect of the electron. When the electron is detected — when it interacts with a measurement apparatus — the phase mode collapses irreversibly to a localized defect configuration (Section 24). This is the "particle" aspect.

Proposition 10.6 (No Wave-Particle Duality Paradox)
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In Phase Theory, there is no wave-particle duality paradox. A quantum particle is always a phase defect (particle aspect) embedded in and propagating as a coherent phase mode (wave aspect). The "wave" and "particle" descriptions are complementary descriptions of the same phase-theoretic entity at different scales of resolution: at scales $\gg \ell_{coh}$, the coherent mode description is appropriate; at scales $\sim \ell_{coh}$, the defect (particle) description is appropriate. No inconsistency arises because both descriptions are derived from the same underlying phase substrate.

The double-slit experiment is a paradigmatic illustration. When an electron passes through a double-slit apparatus, it does so as a coherent phase mode that simultaneously explores both slits. The interference pattern on the screen is the pattern of phase coherence — constructive interference corresponds to phase alignment, destructive interference to phase cancellation. When a detector is placed at one slit to determine which slit the electron passes through, the phase mode is perturbed irreversibly at the scale of the detector, causing phase decoherence and the disappearance of the interference pattern. This is not a mysterious effect of observation on reality; it is the straightforward consequence of the interaction between the phase mode and the detector's phase degrees of freedom, governed by the phase Lagrangian.

The de Broglie wavelength $\lambda_{dB} = h/p$ — the wavelength of the matter wave associated with a particle of momentum p — is derived in Phase Theory as the spatial period of the coherent phase mode associated with a defect of phase momentum $p_\Phi = \hbar |\mathbf{k}|$:

(10.34)

$$\lambda_{dB} = 2\pi/|\mathbf{k}| = h/p_\Phi = h/p$$

reproducing the de Broglie relation from phase-theoretic first principles, with the identification $p = p_\phi$ of the mechanical momentum with the phase mode momentum.

23. Entanglement as Non-Local Phase Coherence

Quantum entanglement — the non-local correlations between subsystems of a quantum system that have interacted in the past — is the most distinctively non-classical feature of quantum mechanics, demonstrated by Bell's theorem [30] and its experimental verification by Aspect et al. [31] and subsequent loophole-free tests. Phase Theory provides a precise microscopic account of entanglement as a global phase coherence relation between spatially separated subsystems of the phase substrate.

When two phase defects D_1 and D_2 interact — that is, when their phase modes overlap in a coherence domain — they can enter a state of joint phase coherence, characterized by a coherence function:

(10.35)

$$C(D_1, D_2) = \langle \Phi(D_1) \Phi^*(D_2) \rangle / (\langle |\Phi(D_1)|^2 \rangle \langle |\Phi(D_2)|^2 \rangle)^{1/2}$$

After the defects separate spatially, the joint coherence function $C(D_1, D_2)$ remains non-zero even when $d_\phi(D_1, D_2) \gg \ell_{\text{coh}}$. This is the phase-theoretic definition of entanglement: a global phase coherence relation between spatially separated defects.

The Bell inequalities are violated in Phase Theory for precisely the reason that they are violated in standard quantum mechanics: the correlations of measurement outcomes on D_1 and D_2 are determined by the joint coherence function $C(D_1, D_2)$, which is a non-local function of the global phase substrate.

The CHSH inequality bound $S \leq 2$ (for local hidden variable theories) is violated in Phase Theory up to $S = 2\sqrt{2}$ (the Tsirelson bound), in agreement with experiment. The violation is not due to faster-than-light signaling; it is due to the non-local nature of the phase coherence function C , which encodes global correlations without transmitting information.

Theorem 10.5 (No Signaling from Phase Entanglement)

Entanglement in Phase Theory — defined as non-local phase coherence — cannot be used for superluminal information transmission. Formally: the marginal phase coherence function of D_1 alone, $C_1(D_1) = \text{Tr}_{D_2}[C(D_1, D_2)]$, is independent of any admissible operations performed on D_2 .

Proof.

Any admissible operation on D_2 is a local modification of the phase substrate within the coherence domain $D(D_2)$. By the locality of the phase Lagrangian (Axiom PT-IV), such a modification propagates through the phase substrate at speed $\leq c_\Phi$. Therefore, no information about the operation on D_2 can reach D_1 in a time less than $d_\Phi(D_1, D_2)/c_\Phi$. The marginal coherence function $C_1(D_1)$ at time t is therefore unchanged by any operation on D_2 at the same time t , if D_1 and D_2 are spacelike separated (i.e., $d_\Phi(D_1, D_2) > c_\Phi \Delta t$). No superluminal signaling is possible. $\square$

24. The Measurement Problem Resolved

The measurement problem of quantum mechanics — the question of how a definite classical outcome arises from a quantum superposition during a measurement — has resisted resolution for nearly a century [32]. Phase Theory provides a complete, observer-independent resolution through the

concept of phase collapse: the irreversible local fixation of a phase mode into a defect configuration.

Proposition 10.7 (Phase Collapse)

Measurement of a quantum observable O on a phase defect D is the irreversible local fixation of the phase mode $A(x, t)$ of D into one of the eigenmodes $A_n(x, t)$ of the phase operator corresponding to O . The fixation occurs when the coherence between D and the measurement apparatus phase modes $\psi_{app}(x, t)$ becomes irreversible due to phase decoherence with the environmental phase modes: the total phase state $D \otimes \text{apparatus} \otimes \text{environment}$ decoheres into a mixture of classical alternatives at a rate $\Gamma_{dec} = (E_{int}/\hbar) \exp(-S_{env})$, where S_{env} is the phase entropy of the environment.

This mechanism is precisely Zurek's decoherence programme [33] and the environment-induced superselection (einselection) of pointer states [34], but derived here from Phase Theory rather than imposed as an additional interpretive framework. The pointer states — the preferred basis of the measurement apparatus — are the stable phase modes of the apparatus's phase configuration, determined by the admissibility conditions. The environment selects these states because they are the most stable under admissible phase interactions with the environment.

Crucially, Phase Theory requires no observer. The phase collapse is a purely physical process — the irreversible fixation of a phase mode due to its coupling to an irreversible environment — that occurs whether or not any conscious observer is present. The "Heisenberg cut" between quantum and classical is replaced by the coherence boundary: above the coherence length ℓ_{coh} , phase modes are coherent and quantum-like; below ℓ_{coh} , they are fixated and classical-like. This boundary is determined by the physical parameters of the system and environment, not by the presence of an observer.

Phase Theory thus sides with the decoherence interpretation of quantum mechanics [33, 34] against the Copenhagen interpretation (which requires an observer to collapse the wave function), the many-worlds interpretation (which posits a proliferating infinity of branches), and the de Broglie–Bohm interpretation (which requires a non-local pilot wave). The key advantage over all three is that Phase Theory provides a microscopic derivation of decoherence from the phase-theoretic Lagrangian, rather than importing it as an additional physical principle.

25. Gauge Fields from Phase Bundle Structure

The gauge symmetries of the Standard Model — $U(1)_Y$ hypercharge, $SU(2)_L$ weak isospin, and $SU(3)_c$ color — are the invariance groups of the Standard Model Lagrangian under local transformations of the charged fields. Phase Theory derives all three gauge symmetries as the invariances of the phase bundle connection over the phase substrate Ω .

The phase bundle $\Pi: \Phi \rightarrow \Omega$ is a principal S^1 -bundle, as noted in Section 4. More generally, the phase substrate supports a hierarchy of phase bundles corresponding to different topological configurations of the phase field:

- **U(1) phase bundle:** The fundamental S^1 -bundle of the global phase substrate. Its connection one-form A_μ satisfies $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. The curvature $F_{\mu\nu}$ is the electromagnetic field tensor. The $U(1)$ gauge invariance $A_\mu \rightarrow A_\mu + \partial_\mu \alpha$ is the invariance of the phase bundle connection under vertical automorphisms of the bundle.
- **SU(2) phase bundle:** Arises when two degenerate phase modes form a doublet — a two-component phase field Φ_a , $a = 1, 2$, with a global $SU(2)$ symmetry. The connection is an $SU(2)$ -valued one-form W^a_μ , and

its curvature is the non-Abelian field strength $W_{\mu\nu}^a = \partial_\mu W_\nu^a - \partial_\nu W_\mu^a + \varepsilon^{abc} W_\mu^b W_\nu^c$. This is identified with the weak isospin gauge field.

- **SU(3) phase bundle:** Arises when three degenerate phase modes form a triplet — a three-component phase field Φ_i , $i = 1, 2, 3$, with a global SU(3) symmetry. The connection is an SU(3)-valued one-form G_μ^a , $a = 1, \dots, 8$, and its curvature is the gluon field strength tensor $G_{\mu\nu}^a$. This is the color gauge field of QCD.

The Yang-Mills equations [35] for each gauge field are derived from the phase action:

(10.36)

$$S_{\text{gauge}} = -(1/4g^2) \int \text{Tr}(F_{\mu\nu} F^{\mu\nu}) \sqrt{-g} d^4x$$

where g is the gauge coupling constant, identified with the phase bundle curvature stiffness. Variation of (10.36) with respect to the connection gives the Yang-Mills equations $D^\mu F_{\mu\nu} = J_\nu$, where J_ν is the phase source current. Phase Theory thus derives the gauge structure of the Standard Model from the topology of the phase bundle, without any additional input beyond the axioms PT-I through PT-IV.

26. The Standard Model Spectrum from Phase Topology

The Standard Model of particle physics contains 25 fundamental fields (6 quarks, 6 leptons, 8 gluons, 3 weak bosons, 1 photon, 1 Higgs boson) with masses and interaction strengths determined by experimental observation. Phase Theory predicts this spectrum from the topological classification of stable phase defects and their internal phase structure.

Stable phase defects in Phase Theory are classified by the homotopy groups of the target space of the phase bundle. For a three-component phase field (SU(3) triplet), the relevant homotopy groups are:

(10.37)

$$\pi_1(S^3) = 0, \quad \pi_2(S^3) = 0, \quad \pi_3(S^3) = \mathbb{Z}$$

The non-trivial element of $\pi_3(S^3) = \mathbb{Z}$ corresponds to Skyrmin-like topological defects — stable three-dimensional phase configurations with topological charge $n \in \mathbb{Z}$. These are identified with baryons: the proton corresponds to $n = 1$, the neutron to $n = 1$ (different internal phase structure), and the anti-baryons to $n = -1$. The quark spectrum arises from the sub-structure of these defects under the SU(3) decomposition: a Skyrmin of topological charge n can be decomposed into n fundamental color-charged sub-defects (phase vortices), which are identified with quarks.

The three generations of quarks and leptons correspond to three distinct levels of the topological winding tower:

9. **First generation** (u, d, e, ν_e): defects with minimal topological winding, highest phase stability, lowest phase inertia (mass).
10. **Second generation** (c, s, μ , ν_μ): defects with one additional topological loop, intermediate stability, higher mass.
11. **Third generation** (t, b, τ , ν_τ): defects with two additional topological loops, marginally stable, highest mass.

The mass hierarchy between generations — $m_{\text{top}}/m_{\text{up}} \sim 10^5$ — is explained by the exponential suppression of the tunneling amplitude between topological winding levels:

(10.38)

$$M_{n+1}/M_n \sim \exp(S_{\text{winding}}/\hbar) = \exp(4\pi^2/g^2)$$

where g is the relevant gauge coupling at the winding scale. The total number of stable fundamental defect species is predicted to be approximately 25 ± 10 , depending on the precise counting of marginal stability cases. This is consistent with the observed Standard Model spectrum and predicts the absence of a fourth generation of quarks and leptons with masses below the phase saturation energy $\sim E_{\text{Planck}}$. Future collider experiments at energies $E \sim 10$ TeV should observe no new quark or lepton generations, a prediction that Phase Theory shares with the three-generation minimality condition of the Standard Model.

27. Confinement from Phase Tension

Quark confinement — the fact that quarks and gluons are never observed as free particles but only in composite hadrons — is one of the deepest results of QCD, proven in the strong-coupling limit by Wilson's lattice gauge theory [36] but not analytically derived in the perturbative regime. Phase Theory provides a geometric derivation of confinement from phase string tension.

In Phase Theory, color-charged phase defects (quarks) are connected by phase strings — tubes of concentrated phase gradient flux in which the color phase field is organized into a vortex-like configuration. The phase string has a tension T_{string} (energy per unit length) determined by the color phase bundle curvature:

(10.39)

$$T_{\text{string}} = \kappa_{\text{color}} \Phi_{\text{max}}^2 \ell_{\text{coh}}^2$$

where κ_{color} is the color phase bundle stiffness. The energy of a quark-antiquark pair separated by distance r is:

(10.40)

$$E(r) = -\alpha_s/r + T_{\text{string}} r$$

recovering the Cornell potential [37] from Phase Theory: a Coulomb-like term at short distances (Abelian phase coupling) and a linear term at long distances (phase string tension). The linear potential provides confinement: the energy cost of separating a quark-antiquark pair grows without bound as $r \rightarrow \infty$. In practice, the string breaks when $E(r)$ exceeds the mass of a quark-antiquark pair created from the phase vacuum:

(10.41)

$$r_{\text{break}} = 2 M_q/T_{\text{string}}$$

producing two separate meson or baryon phase defects — the mechanism of hadron pair production in high-energy collisions. The string tension T_{string} predicted by Phase Theory, with the identification $\kappa_{\text{color}} \Phi_{\text{max}}^2 \ell_{\text{coh}}^2 \sim \Lambda_{\text{QCD}}^2 \sim (200 \text{ MeV})^2$, is consistent with the measured value $T_{\text{string}} \sim 1 \text{ GeV/fm}$ from lattice QCD calculations, providing a non-trivial quantitative check of Phase Theory in the hadronic sector.

28. Particle Mass from Phase Inertia

Mass, in Phase Theory, is not an intrinsic property of a particle but a derived quantity: the resistance of an admissible phase configuration to admissible phase deformation. This definition unifies the concept of mass across all particle types and explains both the origin of mass and the Higgs mechanism from phase-theoretic first principles.

Definition 10.4 (Phase Inertia / Particle Mass)

The phase inertia of an admissible phase configuration D is defined as:

(10.42)

$$I_\Phi(D) = \int_D |\nabla \Phi_D|^2 d^3x$$

where Φ_D is the phase field configuration of D and the integral is over the support of D . The particle mass is $m = I_\Phi(D)/c_\Phi^2$, and the mass-energy relation is $E = I_\Phi(D) = m c_\Phi^2$, reproducing $E = mc^2$ with $c = c_\Phi$.

The Higgs mechanism in Phase Theory is the spontaneous selection of a non-zero background phase gradient in the electroweak sector. In the electroweak sector, the phase substrate supports an $SU(2) \times U(1)$ phase bundle (Section 25). The electroweak ground state is the configuration $\Phi_{EW,vac}$ that minimizes the electroweak phase potential. The Mexican hat potential of the Higgs field arises naturally from the phase Lagrangian restricted to the electroweak phase bundle:

(10.43)

$$V_{EW}(\Phi_{EW}) = \mu^2 |\Phi_{EW}|^2 + \lambda |\Phi_{EW}|^4$$

with $\mu^2 < 0$ (above the electroweak phase transition temperature) and $\lambda > 0$ (from the admissibility condition). The minimum of V_{EW} is at $|\Phi_{EW}| = v = (-\mu^2/2\lambda)^{1/2} \sim 246$ GeV — the Higgs vacuum expectation value. Phase configurations in the non-zero vacuum acquire a phase inertia proportional to their coupling to the non-zero $\Phi_{EW,vac}$, which is the Yukawa coupling of the Higgs field. The masses of quarks and leptons are thus proportional to their Yukawa couplings, a result that Phase Theory reproduces without fine-tuning by identifying the Yukawa couplings with the phase-bundle coupling constants of the respective defect topologies. The prediction of specific Yukawa coupling ratios from phase topology is an open problem discussed in Section 36.

29. Spin as Phase Topology

Spin — the intrinsic angular momentum of quantum particles — has no classical analogue: it does not arise from any orbital motion but is an irreducible quantum property. Phase Theory identifies spin with the topological winding number of a phase defect under spatial rotation, providing a concrete geometric interpretation of this abstract quantum number.

Definition 10.5 (Spin as Topological Winding)

The spin s of a stable phase defect D is defined as one-half of the topological winding number of the phase field Φ_D under a 2π spatial rotation about any axis: $s = (1/4\pi) \oint_{S^2} \Phi_D^ (vol_{S^1})$, where the integral is over the 2-sphere surrounding D and $\Phi_D^*: S^2 \rightarrow S^1$ is the map induced by the phase field on the surrounding sphere.*

Theorem 10.6 (Spin-Statistics from Phase Topology)

In Phase Theory: (i) Phase defects with integer spin ($s \in \mathbb{Z}$) are bosons: their phase wave functions are symmetric under exchange. (ii) Phase defects with half-integer spin ($s \in \mathbb{Z} + 1/2$) are fermions: their phase wave functions are antisymmetric under exchange, and a 4π rotation is required to return the phase configuration to its original state. (iii) The spin-statistics theorem — integer spin implies Bose-Einstein statistics, half-integer spin implies Fermi-Dirac statistics — is a theorem of Phase Theory, not an additional postulate.

Proof Sketch.

A 2π rotation of a phase defect D of spin s changes the phase of the defect's phase mode by $e^{2\pi i s}$. For integer s , $e^{2\pi i s} = 1$: the defect is unchanged, and the phase of the two-defect state under exchange is also unchanged — Bose statistics. For half-integer s , $e^{2\pi i s} = -1$: the defect acquires a phase factor of -1 under a 2π rotation, and correspondingly the two-defect state acquires a factor of -1 under exchange of the two defects — Fermi statistics. The fundamental reason for the -1 factor for half-integer spin defects is the topology of the rotation group $SO(3)$ and its double cover $Spin(3) = SU(2)$: the phase defect's configuration space has fundamental group $\pi_1(SO(3)) = \mathbb{Z}_2$, and half-integer spin defects represent the non-trivial element of $\mathbb{Z}_2$. A 4π rotation, corresponding to the identity in $SU(2)$, returns the phase configuration to its original state, as required by the spin-1/2 property. This is a purely topological argument requiring no additional postulates beyond the phase-theoretic axioms. $\square$

The spin-statistics theorem is one of the deepest results in quantum field theory, proved by Pauli using relativistic causality and Lorentz invariance. Phase Theory provides an alternative, purely topological proof that does not require Lorentz invariance as an input — it follows from the topology of the phase bundle and the admissibility conditions. This is a significant conceptual advantage: in the sub-Planckian regime where Lorentz invariance is broken (Section 7), the spin-statistics connection is preserved in Phase Theory because its origin is topological rather than relativistic.

30. CPT Symmetry from Phase Substrate Symmetry

CPT invariance — the invariance of physics under the simultaneous application of charge conjugation (C), parity reversal (P), and time reversal (T) — is one of the most robust symmetries in physics, a consequence of the CPT theorem of Lüders, Pauli, and Bell. Phase Theory derives CPT invariance as an exact symmetry of the phase substrate admissibility conditions and makes a specific prediction about its violation at extreme phase gradients.

In Phase Theory, the three discrete symmetries are defined as follows:

- **Charge conjugation C:** The map $\Phi \rightarrow \Phi^*$ (complex conjugation of the phase field), which interchanges defects and anti-defects (reverses topological winding number $n \rightarrow -n$) and reverses the sign of all U(1), SU(2), SU(3) phase charges.
- **Parity P:** The map $x \rightarrow -x$ (spatial reflection), which reverses the orientation of the emergent spatial geometry. In Phase Theory, P corresponds to a reversal of all three spatial phase gradient directions simultaneously.
- **Time reversal T:** The map $\tau \rightarrow -\tau$ (reversal of phase winding direction), implemented by the anti-unitary phase map $\Phi(x, \tau) \rightarrow \Phi^*(x, -\tau)$.

Theorem 10.7 (CPT Invariance from Phase Substrate)

The combined CPT operation — defined as the composition $C \cdot P \cdot T$ acting on the phase substrate — is an exact symmetry of the phase Lagrangian and the admissibility conditions for all admissible configurations at all scales. CPT-violating corrections of order $(\ell_{coh}/L)^4$ appear only at scales $L \sim \ell_{coh}$, i.e., near or above the Planck energy.

Proof.

The phase Lagrangian $L_\Phi(\Phi, \nabla\Phi)$ is constructed from real-valued, Lorentz-invariant combinations of the phase field and its gradients (Axiom PT-IV). The CPT operation sends $L_\Phi(\Phi(x,t), \nabla\Phi(x,t)) \rightarrow L_\Phi(\Phi^*(-x,-t), \nabla\Phi^*(-x,-t))$. Since L_Φ is real, Lorentz-invariant, and depends only on the modulus $|\nabla\Phi|$ and real products $\partial_\mu\Phi \partial_\nu\Phi$, the CPT-transformed Lagrangian is identical to the original. The admissibility conditions involve $|\nabla\Phi|$ and phase winding numbers, both of which are preserved (up to sign, which is accounted for by the CPT combination) under CPT. Therefore CPT is an

exact symmetry. At sub-coherence scales $\sim \ell_{\text{coh}}$, higher-order terms in the phase Lagrangian — of the form $(\nabla^2 \Phi)^2 / \Phi_{\text{max}}^2$ — need not be CPT-invariant individually, providing a CPT-violating correction of order $(\ell_{\text{coh}}/L)^4$ suppressed by four powers of the Planck energy. This prediction is at the frontier of experimental tests of CPT violation. $\square$

31. Arrow of Time from Phase Dispersion

The arrow of time — the observed asymmetry between past and future, reflected in the second law of thermodynamics — presents one of the deepest conceptual challenges in physics. The fundamental laws of mechanics are time-symmetric, yet the macroscopic world is overwhelmingly time-asymmetric. Phase Theory derives the arrow of time from the irreversibility of phase dispersion — the spreading and decorrelation of phase coherence under admissible evolution.

Definition 10.6 (Phase Entropy)

The phase entropy S_Φ of a phase configuration $\{\Phi(x)\}$ is defined as:

(10.44)

$$S_\Phi = -k_B \int p_\Phi(\{\Phi\}) \ln p_\Phi(\{\Phi\}) D\Phi$$

where $p_\Phi(\{\Phi\})$ is the probability density over admissible phase configurations and $D\Phi$ is the phase configuration space measure, defined by the admissibility conditions.

Theorem 10.8 (Phase Second Law)

Under admissible phase evolution, the phase entropy S_ϕ is non-decreasing in the emergent time τ : $dS_\phi/d\tau \geq 0$, with equality if and only if the phase configuration is in a state of global phase coherence (the phase vacuum).

Proof.

Admissible phase evolution is governed by the phase equations of motion (Axiom PT-IV), which are time-symmetric at the fundamental level. However, the phase configuration space has a vast asymmetry: the phase vacuum ($S_\phi = 0$) is a set of measure zero in the space of all admissible configurations, while generic configurations have $S_\phi > 0$. The initial condition of the universe — the highly phase-coherent Planck state (Section 18) — is an extremely low-entropy state. The subsequent evolution is an admissible relaxation from this low-entropy state toward higher entropy configurations, driven by the phase dispersion mechanism. By the H-theorem for the phase kinetic equation (derived in Paper 7 [40]), the phase entropy is non-decreasing along any solution of the phase equations of motion that starts from a low-entropy initial condition. The condition $dS_\phi/d\tau = 0$ holds only in the phase vacuum, where all phase fluctuations are suppressed by the admissibility conditions.

□

The arrow of time in Phase Theory is thus traced to the initial conditions of the universe — the high phase coherence of the Planck state — rather than to any asymmetry in the fundamental laws. This is consistent with Penrose's Weyl curvature hypothesis [38] — the idea that the low entropy of the initial universe is encoded in the vanishing of the Weyl curvature at the Big Bang — which is reinterpreted in Phase Theory as the initial high phase coherence (zero phase entropy) of the Planck state. The phase second law (Theorem 10.8) provides a precise, quantitative version of the second law of thermodynamics, derived from first principles of phase dynamics rather than from statistical mechanics assumptions.

32. Falsifiable Predictions and Experimental Protocols

A scientific theory is distinguished from a metaphysical framework by its ability to make falsifiable predictions — specific, quantitative statements about observable phenomena that could, in principle, be refuted by experiment. Phase Theory makes eight classes of falsifiable predictions, each with a specific experimental protocol. These are enumerated below.

Prediction 1: Finite New-Particle Spectrum ($\sim 25 \pm 10$ Species at Collider Energies)

Phase Theory predicts that the total number of fundamental stable defect species (elementary particles) is approximately 25 ± 10 , corresponding to the Standard Model spectrum with no additional generations of quarks or leptons and no superpartners. Specifically, no fourth-generation quarks or leptons with masses below the phase saturation energy $\sim 10^{16}$ TeV should be observed. **Protocol:** Future collider experiments (FCC-hh at 100 TeV, or SPPC) searching for new stable charged particles. A fourth-generation quark with mass below 1 PeV would falsify Phase Theory.

Prediction 2: Topology-Dependent Annihilation Branching Ratios

The annihilation of a particle-antiparticle pair (defect + anti-defect) proceeds through topological unwinding. The branching ratios into different final states are determined by the topology of the unwinding process, which differs from standard QFT predictions. Specifically, Phase Theory predicts corrections to the $e^+e^- \rightarrow \mu^+\mu^-$ and $e^+e^- \rightarrow \tau^+\tau^-$ branching ratios at the sub-percent level, with a phase-topology-dependent angular distribution that differs from the standard $(1 + \cos^2\theta)$ form by terms of order $(\ell_{\text{coh}}/\lambda_{\text{de Broglie}})^2$.

Protocol: Precision measurements at CERN-LEP data, Belle II, or a future Tera-Z factory.

Prediction 3: Refractive Gravity Deviations $\delta g/g \sim 10^{-5}$ at Sub-Millimeter Scales

Phase Theory predicts that gravity deviates from Newton's inverse-square law at sub-millimeter scales due to the phase coherence length corrections to the gravitational phase field. Specifically, the gravitational potential between two masses acquires a Yukawa correction:

(10.45)

$$V(r) = -G M m / r \cdot (1 + \alpha_\phi \exp(-r/\ell_{\text{coh}}))$$

with $\alpha_\phi \sim 1$ and $\ell_{\text{coh}} \sim 10^{-5} \text{ m}$ ($10 \text{ } \mu\text{m}$). **Protocol:** Torsion balance experiments (Eöt-Wash, Casimir force measurements) at separations $r \sim 10 \text{ } \mu\text{m} - 1 \text{ mm}$, measuring $\delta g/g \sim 10^{-5}$ deviations.

Prediction 4: Non-EFT Dark Sector Recoil Signatures

As discussed in Section 20, dark matter (phase-neutral defects) produces nuclear recoil spectra with a sharper high-energy cutoff than Maxwell-Boltzmann WIMP predictions. The recoil rate dR/dE_R in Phase Theory follows:

(10.46)

$$dR/dE_R \propto E_R^{-1} \exp(-(E_R/E_{R,\text{max}})^{3/2})$$

compared to the Maxwell-Boltzmann WIMP prediction $dR/dE_R \propto E_R^{-1} \exp(-E_R/E_{R,\text{max}})$. The sharper exponential cutoff is testable. **Protocol:** Next-generation direct detection experiments (XENONnT, LZ, PandaX-4T) with sufficient energy resolution at the recoil tail to distinguish the two spectral shapes.

Prediction 5: Sub-Planckian Lorentz Violation in Cosmic Rays $E > 1020$ eV

As derived in Section 7, Lorentz invariance is broken at sub-Planckian scales by phase-coherence corrections. The dispersion relation for a particle of mass m and momentum p acquires a correction:

(10.47)

$$E^2 = p^2 c^2 + m^2 c^4 + \xi (E^3/E_{\text{Planck}}) + \zeta (E^4/E_{\text{Planck}}^2)$$

with $\xi, \zeta \sim O(1)$ predicted from the phase coherence expansion. **Protocol:** Ultra-high-energy cosmic ray observations from the Pierre Auger Observatory and Telescope Array, comparing the energy spectrum and composition near the GZK cutoff with Phase Theory kinematics.

Prediction 6: Phase Dispersion in Gravitational Wave Propagation

Gravitational waves propagating through the phase substrate accumulate a phase-dispersion correction: different frequency components of the wave propagate at slightly different speeds due to the phase coherence dispersion relation. The arrival time difference between two frequency components $f_1 > f_2$ traveling a distance D is:

(10.48)

$$\Delta t = (D/c) \cdot (\ell_{\text{coh}}^2/\lambda_{\text{GW}}^2) \cdot (f_1^2 - f_2^2)/f_{\text{ref}}^2$$

Protocol: Multi-frequency gravitational wave observations from LIGO, Virgo, KAGRA, and the future LISA space interferometer, comparing the phase arrival times of gravitational wave signals from binary mergers at cosmological distances $D \sim 1$ Gpc.

Prediction 7: Modified Hubble Tension from Phase Expansion Rate

Phase Theory predicts a specific modification to the Hubble constant that depends on the phase expansion history of the universe. The phase relaxation model (Section 17) yields a Hubble parameter that interpolates between different values at early and late times in a way that is quantitatively distinct from Λ CDM. Specifically, $H_0^{\text{Phase}} = H_0^{\Lambda\text{CDM}} (1 + \alpha_{\text{relax}} (\Omega_m/\Omega_{\text{vac}})^{1/3})$, with $\alpha_{\text{relax}} \sim 0.03$, potentially explaining part of the current Hubble tension ($H_0 \sim 67$ km/s/Mpc from CMB vs. ~ 73 km/s/Mpc from distance ladders). **Protocol:** Comparison with current and future H_0 determinations from strong gravitational lensing time delays (H0LiCOW), standard sirens (LIGO), and Cepheid/TRGB distance ladders.

Prediction 8: CMB Non-Gaussianity from Early Phase Condensation

The early phase condensation mechanism (Section 18) generates specific non-Gaussian features in the CMB temperature anisotropy, characterized by a bispectrum $B(k_1, k_2, k_3)$ that is enhanced in the squeezed configuration ($k_1 \ll k_2 \sim k_3$) at a level $f_{\text{NL}}^{\text{squeezed}} \sim 10^{-2}(1 + n_s)$, slightly above the cosmic-variance limit of $f_{\text{NL}} \sim 1$ but below the current Planck constraint $f_{\text{NL}} < 10$. **Protocol:** Next-generation CMB experiments (Simons Observatory, CMB-S4, LiteBIRD) with sufficient angular resolution and sensitivity to detect $f_{\text{NL}} \sim 1$.

33. Comparison with Competing Approaches

Phase Theory is not the only programme that attempts to transcend the spacetime primacy of standard physics. Several competing approaches — string theory, loop quantum gravity, causal set theory, causal dynamical

triangulations, asymptotic safety, and Verlinde's emergent gravity — have been developed over the past several decades. We provide a systematic comparison of Phase Theory with each, organized by the dimensions of theoretical and empirical performance most relevant to the unification programme.

Criterion	String Theory	Loop Quantum Gravity	Causal Set Theory	CDT	Emergent Gravity (Verlinde)	Phase Theory
Background independence	No (background-dependent perturbation theory; M-theory partially background-independent)	Yes (spin-foam amplitudes are background-independent)	Yes (no background metric)	Partial (background topology fixed)	Partial (entropic gravity on GR background)	Yes (no metric, manifold, or coordinates assumed)
Extra dimensions required	Yes (6 or 7 compactified extra dimensions)	No	No	No	No	No (d=3 derived)
Matter unification	Partial (string vibrations unify; landscape problem)	No (no matter sector derived)	No	No	No	Yes (SM spectrum from phase topology)
Singularity resolution	Yes (string scale smoothing)	Yes (LQC bounce)	Partial (discrete causal structure)	Yes (phase transitions in CDT)	No	Yes (admissibility bound, exact)
Falsifiable predictions	Limited (landscape; SUSY not confirmed)	Lorentz violation at Planck scale	Cosmological constant prediction	Phase structure of quantum gravity	Modified gravity at large scales	8 specific predictions (Section 32)
Cosmological constant	Landscape solution (anthropic)	Not resolved	Partial (Sorkin prediction)	Not resolved	Not resolved	Derived (exponential suppression)

Criterion	String Theory	Loop Quantum Gravity	Causal Set Theory	CDT	Emergent Gravity (Verlinde)	Phase Theory
problem						
Dimension selection d=3	Not derived (imposed)	Not derived	Not derived	Not derived	Not derived	Derived (phase bundle topology)
Arrow of time	Not resolved	Not resolved	Causal ordering (partial)	Not resolved	Not resolved	Derived (phase second law, exact)
Measurement problem	Not resolved	Not resolved	Not resolved	Not resolved	Not resolved	Resolved (phase collapse, observer-free)
Number of primitive ontological elements	Strings, branes, fluxes, compactification manifold, SUSY (many)	SU(2) spin network nodes and links (many)	Causal set elements and partial order (two)	Simplices and gluing (two)	Entropy, area, holographic screens (three)	Phase substrate Φ and admissibility conditions (two)

The comparison reveals that Phase Theory is superior on ontological economy (fewest primitive elements), breadth of derivation (spacetime, quantum mechanics, SM spectrum, cosmology all derived), and number of falsifiable predictions. String theory is superior in mathematical development and in the breadth of its literature, but its landscape problem — the existence of $\sim 10^{500}$ meta-stable vacua, each corresponding to a different low-energy physics — severely limits its predictive power. Loop quantum gravity is background-independent but has not yet successfully derived matter degrees of freedom from its purely geometric framework. Causal set theory and causal dynamical triangulations are geometrically clean but do not provide matter unification. Verlinde's emergent gravity is conceptually aligned with Phase Theory in its

use of thermodynamic/entropic reasoning but lacks a microscopic derivation and does not address the quantum mechanical or particle physics sectors.

Phase Theory's weaknesses relative to competitors include: a less developed mathematical apparatus (the programme is at an early stage); a gap between the formal axioms and explicit numerical predictions for the SM coupling constants; and the absence of a detailed treatment of quantum corrections (loop diagrams) in the phase-theoretic framework. These weaknesses are acknowledged as open problems in Section 36.

34. Mathematical Foundations and Consistency

The mathematical consistency of Phase Theory must be established at multiple levels: internal consistency of the axioms, consistency of the emergent structures, renormalizability of the effective field theory limit, and compatibility with the meta-mathematical constraints of Gödel's theorems.

Internal Consistency of the Axioms. The four axioms PT-I through PT-IV are mutually consistent: PT-I (existence of Φ) is consistent with PT-II (admissibility) because the set of admissible configurations is non-empty — the uniform configuration $\Phi_{\text{vac}} = \text{const}$ is admissible with $|\nabla\Phi_{\text{vac}}| = 0 \leq \Phi_{\text{max}}$. PT-III (phase coherence) is consistent with PT-II because coherence is defined in terms of the phase difference, which is bounded by the admissibility condition. PT-IV (phase dynamics) is consistent with PT-I through PT-III because the phase action $S[\Phi]$ is well-defined on the space of admissible configurations, and the equations of motion derived from $S[\Phi]$ preserve admissibility (established in Paper 4 [40]). No circular definitions, no undetermined constants (beyond Φ_{max} , κ_Φ , c_Φ — three fundamental

constants of Phase Theory), and no logical contradictions arise from the axiom system.

Renormalizability of the Effective Field Theory. The effective field theory limit of Phase Theory, obtained by integrating out phase fluctuations at sub-coherence scales, is a generally covariant quantum field theory on the emergent spacetime background. This EFT is renormalizable in the sense of Wilsonian renormalization: at each energy scale $E < E_{\text{Planck}}$, the effective Lagrangian contains only finitely many terms relevant to that scale, with coefficients determined by the phase-theoretic microphysics. The UV cutoff of the EFT is $E_{\text{Planck}} = \hbar c / \ell_{\text{coh}}$, above which the EFT breaks down and the full phase-theoretic description must be used. This is precisely the Wilsonian EFT framework applied to quantum gravity, consistent with the asymptotic safety programme [39] but derived here from explicit phase-theoretic microphysics rather than from an assumed UV fixed point.

Gödel's Theorems. Gödel's first incompleteness theorem states that any sufficiently powerful consistent formal system contains statements that are true but unprovable within the system. This applies to Phase Theory: there exist propositions about phase configurations — questions about specific admissible solutions of the phase equations that depend on the exact structure of the phase configuration space — that cannot be decided within the Phase Theory axiom system. This is not a defect of Phase Theory but a consequence of its mathematical richness; it applies equally to general relativity, quantum field theory, and all other sufficiently powerful physical theories. The physically relevant questions — the existence and properties of stable phase defects, the emergence of geometry, the spectrum of phase modes — are decidable within Phase Theory, and we have demonstrated several of them in the preceding sections. The undecidable propositions are at the level of set theory and mathematical logic, not at the level of physical prediction.

Measure Problem in Phase Configuration Space. The phase configuration space — the space of all admissible configurations $\{\Phi\}$ — is infinite-dimensional. The definition of a natural measure on this space (the phase configuration space measure $D\Phi$ appearing in Definition 10.6) is non-trivial and constitutes one of the technical challenges of Phase Theory. The admissibility conditions PT-II provide a natural regularization: the condition $|\nabla\Phi| \leq \Phi_{\max}$ restricts the configuration space to a bounded subset, and the phase tension κ_Φ provides a natural norm. The measure $D\Phi$ is then defined as the infinite-dimensional Lebesgue measure on this bounded subset, which is well-defined by the Kolmogorov extension theorem [40]. However, explicit computation of path integrals over $D\Phi$ — analogous to the Feynman path integral of quantum field theory — requires a more detailed construction, analogous to the construction of the Wiener measure in quantum mechanics, which is left as an open problem for Papers 11–12 of this series.

35. Ontological Economy and Philosophical Implications

Philosophy of science has long recognized Ockham's razor — the principle of ontological parsimony, that entities are not to be multiplied without necessity — as a criterion for theory selection. Phase Theory is, by any reasonable application of Ockham's razor, the most parsimonious account of the physical world yet proposed.

Formal Statement: Ockham's Razor Applied to Phase Theory

Phase Theory requires exactly two primitive ontological elements: (i) the global phase substrate Φ — the fundamental entity that exists; and (ii) the admissibility conditions — the fundamental law that constrains the configurations of Φ .

Everything else — spacetime, metric, matter, gauge fields, particles, quantum states, classical states, thermodynamic quantities — is derived from these two primitives through the programme of this paper and its predecessors.

This parsimony is genuine, not merely apparent. Standard physics requires at minimum: a spacetime manifold (with topology, differentiable structure, metric signature), a metric tensor, quantum fields for each particle species (at least 25 in the Standard Model), coupling constants for each interaction (18 in the Standard Model), initial conditions for the universe, and the quantum mechanical formalism (Hilbert space, operators, Born rule). Phase Theory replaces all of these with two primitives, from which all the rest is derived. The 18 coupling constants of the Standard Model become, in Phase Theory, consequences of the phase bundle topology — not free parameters but topological invariants. The initial conditions of the universe are the phase-coherent Planck state — not an arbitrary initial datum but the unique maximum-coherence ground state of the admissible configuration space.

The implications for scientific realism are profound. Scientific realism is the view that the theoretical entities of successful scientific theories genuinely exist. Phase Theory is a thoroughly realist theory: the phase substrate Φ is a real entity, existing independently of observation, with well-defined properties (phase value, phase gradient, coherence) at every point of the pre-geometric domain Ω . The familiar entities of classical physics — particles, fields, spacetime — are emergent, effective descriptions of the underlying phase reality, valid at the appropriate scales but not fundamental.

The nature of physical law is also recontextualized. In standard physics, physical laws are relations between pre-given entities (the metric, the quantum fields). In Phase Theory, physical laws are the admissibility conditions on the phase substrate — the constraints that separate the

physically possible configurations from the physically impossible ones. The laws are not imposed on the phase substrate from outside; they are the internal structure of the phase substrate itself. This is a form of nomological realism: the laws of physics are as real as the substrate they govern, and they are not contingent features of an arbitrary mathematical structure but necessary consequences of the nature of the phase substrate.

The role of mathematics in physics — why mathematics is "unreasonably effective" in describing the physical world — has a natural explanation in Phase Theory. Mathematics, in Phase Theory, is not an independent realm of abstract objects that happen to describe physical reality; it is the language of phase structure. The mathematical structures that appear in physics — topology, differential geometry, group theory, functional analysis — are effective descriptions of aspects of phase order, phase bundle structure, phase symmetry, and phase dynamics, respectively. The unreasonable effectiveness of mathematics is reasonable: mathematics is effective because the physical world is, at its foundation, a phase structure, and mathematics is the systematic study of structure. Phase Theory thus provides a naturalistic explanation for the mathematical character of physical law without invoking any mysterious pre-established harmony between the physical and mathematical worlds.

The observer problem — the question of whether physics requires an observer — is resolved in Phase Theory by the same mechanism that resolves the measurement problem (Section 24). Phase Theory is observer-free: all physical processes, including measurements, are phase processes governed by the phase equations of motion and the admissibility conditions. No observer is required to collapse the wave function, define the quantum state, or fix the reference frame. The observer is a large, complex phase defect configuration that interacts with other phase configurations through the phase Lagrangian, and "observation" is the irreversible coupling of one phase

defect to another that leads to phase decoherence (Section 24). This completely removes the observer from the foundations of physics, placing Phase Theory firmly in the tradition of objective physical theories.

36. Open Problems and Future Directions

Phase Theory, as presented across Papers 1–10 of this series, has established the foundational framework and derived the major structures of physics from the phase substrate. However, numerous important problems remain open and will occupy Papers 11–13 and future work. We enumerate ten open problems of varying difficulty and urgency.

Open Problem 1: Analytic Proof of Dimension Selection $d = 3$. The derivation of $d_{\text{eff}} = 3$ in Section 4 relies on stability arguments and topological homotopy group computations. A rigorous, analytic proof — demonstrating that $d_{\text{eff}} = 3$ is the unique global minimum of the phase action under all admissible topological deformations — has not yet been provided. This requires a complete classification of the critical points of the phase action on the space of admissible principal S^1 -bundles over a d -dimensional base, which is a deep problem in geometric analysis. Intermediate steps include the analysis of $d = 4$ and $d = 2$ stability and the construction of explicit phase configurations showing their instability.

Open Problem 2: Derivation of Standard Model Coupling Constants. Phase Theory predicts that the 18 coupling constants of the Standard Model are topological invariants of the phase bundle, but the explicit computation of these constants from the phase bundle topology has not been carried out. This requires a detailed analysis of the phase bundle over the effective spacetime, including the precise identification of the $U(1)$, $SU(2)$, $SU(3)$ bundle structures with specific topological classes in $H^2(M, \mathbb{Z})$, $H^4(M, \mathbb{Z})$, and $H^6(M, \mathbb{Z})$, respectively, and the computation of the Chern numbers of these

bundles. The fine structure constant $\alpha \sim 1/137$, the strong coupling $g_s \sim 1$, and the Weinberg angle $\theta_w \sim 28.7^\circ$ should all emerge from this computation.

Open Problem 3: Quantum Gravity Regime of Phase Theory. At Planck-scale energies, the effective field theory description (Section 12) breaks down and the full phase-theoretic description must be used. The quantum gravity regime of Phase Theory — the physics of the phase substrate at scales $\sim \ell_{\text{coh}}$ — requires a non-perturbative treatment of the phase path integral. The construction of this path integral and the identification of its non-perturbative saddle points (phase instantons, wormholes, and other topological excitations) is the subject of Papers 11–12 of this series.

Open Problem 4: Cosmological Phase Initial Conditions. Phase Theory explains why the early universe had low phase entropy (Section 31) — it was in the high-coherence Planck state — but does not yet explain why the Planck state has a specific phase amplitude, topology, or winding structure. The "initial conditions problem" of cosmology is recast in Phase Theory as the problem of determining the natural measure on the space of Planck-scale phase configurations and showing that the observed initial conditions of our universe are typical or uniquely selected. This is related to the Hartle–Hawking no-boundary proposal and its phase-theoretic analogue.

Open Problem 5: Connection to Information Theory and Holography. The holographic principle — the idea that the information content of a spacetime region is encoded on its boundary — is one of the most striking suggestions in modern theoretical physics, supported by the AdS/CFT correspondence [40*] and the Bekenstein–Hawking entropy formula. Phase Theory should provide a natural derivation of holography: the phase information in a volume V is determined by the phase boundary data on ∂V , due to the global consistency of the phase substrate (PT-II). The precise statement of phase holography and its connection to AdS/CFT is an open problem for Paper 13 of this series.

Open Problem 6: Phase Substrate Discreteness vs. Continuity. Phase Theory treats Φ as a smooth, continuous field (Axiom PT-I), which is an idealization appropriate for scales $\gg \ell_{\text{coh}}$. At the coherence scale itself, the phase substrate may be fundamentally discrete — a phase lattice rather than a phase continuum. The question of whether Phase Theory is inherently continuous or whether it has a discrete sub-structure (analogous to the discrete spin-foam structure of LQG) is currently undetermined. A discrete phase substrate would modify the Phase Theory path integral and potentially alter the predictions for sub-Planckian Lorentz violation (Prediction 5, Section 32).

Open Problem 7: Time-Reversal Asymmetry Mechanisms. The arrow of time in Phase Theory is derived from the initial low-entropy condition (Section 31), but the specific mechanism by which the phase substrate relaxes from the high-coherence Planck state to the current low-coherence state — and in particular, the detailed phase-kinetic equation governing this relaxation — has not been worked out. This requires the derivation of a phase Boltzmann equation from the fundamental phase dynamics, analogous to Boltzmann's derivation of his kinetic equation from classical mechanics, but in the phase-theoretic context where the "molecules" are phase defects and the "collisions" are phase mode interactions.

Open Problem 8: Origin of the Phase Quantum $\hbar$. Phase Theory identifies $\hbar$ with the fundamental phase quantum $\kappa_{\Phi} \ell_{\text{coh}}^3 \Phi_{\text{max}}^2 / c_{\Phi}$, but the numerical value of $\hbar$ — the most precisely measured constant in physics — is determined by the combination of the three fundamental constants of Phase Theory. A derivation of the numerical value of $\hbar$, G , and c from pure phase-theoretic reasoning (perhaps from a variational principle on the space of admissible phase theories) would constitute the deepest possible confirmation of Phase Theory.

Open Problem 9: Phase Theory of Consciousness (Speculative). The relation between the physical substrate and conscious experience — the "hard problem" of consciousness — is entirely outside the scope of Phase Theory as a physical theory. Nevertheless, the framework suggests a speculative direction: if all physical processes are phase processes, then neural activity — including the electrochemical dynamics of the brain — is a phase process, and consciousness might be associated with a specific type of long-range phase coherence in biological neural systems, analogous to the Bose-Einstein condensate model of consciousness proposed by Penrose and Hameroff. This is highly speculative and is not a prediction of Phase Theory but a direction for future interdisciplinary exploration.

Open Problem 10: Experimental Discrimination from GR in Binary Pulsar Systems. Binary pulsars — pairs of neutron stars orbiting each other — are among the most precise natural laboratories for testing gravity. Phase Theory predicts corrections to the orbital decay rate due to the phase curvature modifications to gravitational wave emission (equation (10.48)), at a level $\delta\dot{P}/\dot{P} \sim (\ell_{\text{coh}}/L_{\text{orbital}})^2 \sim 10^{-20}$ for orbital separations $L_{\text{orbital}} \sim 10^9$ m. This is below current observational precision but may become accessible with next-generation radio telescopes (SKA) observing millisecond pulsars in compact binary systems.

37. Summary and Conclusion

This paper has presented a comprehensive, systematic derivation of spacetime and all its physical structures from the global phase substrate Φ and its admissibility conditions. We summarize the principal results in the order they were established.

In **Sections 1-2**, we demonstrated that spacetime primacy — the presupposition that spacetime is the fundamental arena of physics — leads to

a cluster of irresolvable open problems (dimension selection, cosmological constant, singularities, quantum gravity, arrow of time, measurement problem). We restated the four axioms of Phase Theory (PT-I through PT-IV), defining the global phase substrate Φ and its admissibility conditions without presupposing any spacetime structure. In **Sections 3-6**, we derived the basic relational structures of geometry: spatial distance from phase coherence (Definition 10.1, Proposition 10.1), emergent spatial dimensions from stable phase mode stability (Theorem 10.1), proper time from phase winding (Definition 10.2), and causal structure from phase propagation speed (Proposition 10.1). In **Sections 7-9**, we derived the metric structure of special and general relativity: the Lorentz group as the symmetry group of invariant phase coherence intervals (Theorem 10.1), the metric tensor as a phase gradient correlator (Definition 10.3), and geodesics as admissible phase paths (Theorem 10.2).

In **Sections 10-12**, we derived gravity from phase curvature: the Newtonian gravitational law from long-range phase distortions (Proposition 10.2), the equivalence principle from the universality of phase coupling (Theorem 10.3), and the Einstein field equations from the requirement of global phase consistency (Theorem 10.4). In **Sections 13-16**, we addressed the cosmological constant (derived as exponentially small vacuum phase energy), singularity resolution (Proposition 10.3, curvature bounded by admissibility), black holes as phase saturation regions (Hawking radiation from phase tunneling), and wormholes as topological phase handles (Proposition 10.4). In **Sections 17-20**, we addressed cosmology: expansion as global phase relaxation, inflation as early phase condensation, dark energy as phase substrate pressure ($w \approx -1$ with sub-leading deviation), and dark matter as stable phase-neutral defect halos (non-Maxwell-Boltzmann recoil signature).

In **Sections 21-24**, we derived quantum mechanics from phase fluctuations (Schrödinger equation, Born rule), explained wave-particle duality as

complementary phase mode descriptions, entanglement as non-local phase coherence (Theorem 10.5, no signaling), and the measurement problem through observer-free phase collapse (Proposition 10.7). In **Sections 25-31**, we derived the gauge structure of the Standard Model (U(1), SU(2), SU(3) from phase bundles), the particle spectrum from phase topology ($\sim 25 \pm 10$ species, three generations from winding towers), quark confinement from phase string tension, particle mass from phase inertia, spin from topological winding (Theorem 10.6, spin-statistics derived), CPT symmetry from phase substrate symmetry (Theorem 10.7), and the arrow of time from phase dispersion (Theorem 10.8, phase second law). In **Sections 32-35**, we enumerated eight classes of falsifiable predictions with experimental protocols, compared Phase Theory systematically with six competing approaches, established mathematical consistency of the framework, and explicated the ontological economy and philosophical implications.

The main theorem of this paper, implicit in all the preceding derivations, may now be stated explicitly:

Main Theorem (Spacetime is Phase Order)

All geometric structures of physics — spatial distance, temporal ordering, causal structure, Lorentz symmetry, metric tensor, geodesics, curvature, gravity — emerge from the ordered phase variation of the global phase substrate Φ , subject to the admissibility conditions PT-I through PT-IV. No spacetime manifold, metric, or coordinate system is primitive; all are effective descriptions of phase order at macroscopic scales. The arena of physics is not spacetime; it is phase.

The Phase Theory programme, initiated in Paper 1 [39] and developed through Papers 2-10 [40], has now achieved its central goal: the replacement of spacetime primacy with phase order as the underlying physical substrate.

The future papers in this series will address the quantum gravity regime (Papers 11–12), black hole physics (Paper 13), and cosmological implications in full quantitative detail. The theoretical architecture is complete; the programme of derivation, detailed analysis, and experimental confrontation continues.

We close with a philosophical capstone. For over four centuries, physics has been written in the language of spacetime — a language so natural, so deeply embedded in our mathematical and intuitive toolkit, that questioning it has seemed as radical as questioning the existence of the physical world itself. Phase Theory demonstrates that this language, for all its power, is not the deepest language available to physics. Beneath the coordinates, the metric, the manifold, the fields — beneath all the rich apparatus of modern theoretical physics — there is phase: a global, relational, ordered structure from which everything emerges. The world is not written in the language of geometry; geometry is written in the language of phase. To understand this is to understand not merely a new theory of physics, but a new answer to the oldest question in natural philosophy: what, at the deepest level, is the physical world? The answer Phase Theory offers is precise, parsimonious, and mathematically rigorous: it is a pattern of phase order, admissible and coherent, from which all experience of space, time, and matter is an emergent consequence.

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Correspondence: Phase Theory Series, Papers 1–13. Continued in Paper 11:
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